UNIVERSITÀ DEGLI STUDI DI BERGAMO
Dipartimento di Scienze Aziendali

Strategic Management Accounting

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The Balanced Scorecard

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Learning Objectives

1. Give examples of measures that fit each of the four scorecard perspectives.
2. Name the four types of quality cost and use them to prepare a quality cost report.
3. Calculate throughput time, delivery cycle time, manufacturing cycle efficiency and overall equipment effectiveness.
4. Construct a balanced scorecard, link its measures and use it to evaluate a strategy.



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Strategy comes first

Section I

The Balanced Scorecard

Enel: a green number that sets the interest rate

In 2019 Enel, Italy's largest energy company, borrowed $1.5 billion with a new kind of bond: the interest it pays depends on a **non-financial measure**.

The measure

Renewable capacity as a share of all its plants

→

The target

At least 55% by the end of 2021 (45.9% in mid 2019)

→

If the target is missed

Interest rate up by 0.25 points until maturity

What happened

Investors ordered about $4 billion of bonds, almost three times the amount offered. A green number had become part of a financial contract.

The idea of this lecture

Non-financial measures matter when they are **linked** to results that stakeholders value. A balanced scorecard is built to show those links.

Companies measure performance for two broad reasons

To inform

Measures give employees the information they need to carry out the strategy. A shift supervisor who sees today's defect rate knows where to act.

They also give managers feedback: is the plan working or not?

To influence

Measures shape rewards such as bonuses and promotions. People pay attention to what they are evaluated on.

This is powerful, and it can backfire if the measure is badly chosen.

Why profit alone is not enough

It arrives late

Profit tells us how the last quarter went. By the time it falls, the causes happened months ago.

It does not say why

A lower margin may come from defects, slow deliveries, lost customers or untrained staff. The number alone does not tell us which.

It can be managed

Cutting training or maintenance raises profit this year and damages the business next year.

Financial results remain the final test. The point is that they need measures that show what is happening **before** the money moves.

Why does a scorecard need non-financial measures?

Your turn

- The balanced scorecard gets its name from the attempt to **balance financial and non-financial measures**.
- The aim is to evaluate both short run and long run performance in a single report.
- The scorecard relies on non-financial measures in addition to financial ones for **two reasons**.

Think for one minute

What can non-financial measures tell managers that profit cannot?

Two reasons for non-financial measures

REASON 1

They capture fundamental changes

Non-financial and operational indicators measure the fundamental changes a company is making.

Financial measures are **lag** indicators: they summarise the results of past actions.

Non-financial measures are **leading** indicators of future financial performance.

REASON 2

They are easier to understand and influence

Top managers are usually responsible for financial measures, not lower level managers.

Non-financial measures are more likely to be understood and controlled by lower level managers.

A warehouse supervisor cannot move return on equity, but can move the percentage of orders shipped on time.

Leading and lagging indicators

- A **lagging indicator** records an outcome that has already happened: profit, sales, market share.
- A **leading indicator** moves earlier and helps predict the outcome: training hours, defect rates, delivery times.
- A good scorecard mixes both, so managers can act on the drivers and still check the results.

Leading	Lagging
Hours of staff training	Errors per order
Errors per order	Customer satisfaction
Customer satisfaction	Repeat purchases
Repeat purchases	Profit

Notice: the same measure can be lagging for one step and leading for the next.

DEFINITION

A balanced scorecard is an integrated set of performance measures that are derived from and support a company's strategy.

Integrated means the measures are linked to each other, not just a list.

Derived from means we start from the strategy and then choose the measures.

Support means the measures help people put the strategy into action.

From mission and strategy to measures

- The balanced scorecard translates an organisation's **mission and strategy** into a comprehensive set of performance measures.
- It does not focus solely on financial objectives. It highlights the **non-financial objectives** an organisation must achieve to meet its financial objectives.



Management translates its strategy into performance measures that employees understand and influence.

Where the idea comes from

1992

Robert Kaplan and David Norton publish an article in the *Harvard Business Review* proposing a balanced set of financial and operational measures.

1996

Their book presents the scorecard as a management system for translating strategy into action, not only as a reporting tool.

2004

They add strategy maps: diagrams that show how the objectives in the four perspectives cause one another.

The word **balanced** refers to balance between financial and non-financial measures, short and long term, results and drivers, internal and external views.

What do we mean by strategy?

Your turn

- A **strategy** is a company's "game plan" for attracting customers by distinguishing itself from competitors.
- The distinguishing features are called the **customer value proposition**: the reason a customer should choose us and not someone else.
- The value proposition is the essence of strategy. If we cannot state it in one sentence, we are not ready to build a scorecard.

QUICK TEST

"You should choose us because ..."

Finish the sentence for a company you know well.

The ending is its value proposition.

“You should choose us because…”

FURNITURE

IKEA

…well designed furniture costs little, because you carry it home and build it yourself.

AIRLINES

Ryanair

…we fly you there for less than anyone else, with no frills.

TECHNOLOGY

Apple

…our products are simple, beautiful and work together seamlessly.

CARS

Ferrari

…you will own a rare, extraordinary car with a racing heritage nobody else can offer.

RETAIL

Amazon

…you find almost anything, at a fair price, delivered to your door in a day or two.

HOTELS

Ritz Carlton

…our staff remember your preferences and are free to fix any problem on the spot.

Same industry, opposite promises

Competing in the same market does not mean making the same promise. Each pair sells similar things to very different customers.

Cars	**Dacia** A new car with everything you need and nothing you do not, at the lowest price.	**Ferrari** Performance and exclusivity, built in limited numbers.
Fashion	**Zara** The latest trends in the shop within weeks, at accessible prices.	**Brunello Cucinelli** Handmade cashmere that lasts for years, made in the village of Solomeo.
Groceries	**Lidl** Low prices on a short, carefully chosen range of products.	**Eataly** High quality Italian food, with tastings and cooking classes in the store.

Different promises need **different measures**: that is why two companies in the same industry should not share a scorecard.

Strategy matches capabilities with opportunities

A strategy describes how an organisation matches its **capabilities** with the **opportunities** in the marketplace to accomplish its overall objectives.

In formulating its strategy, an organisation must understand two things:

The industry in which it operates

Customers, competitors, suppliers, substitutes and the rules of the game.

Its strategic resources

What it owns or knows that others cannot easily copy: people, brands, technology, location.

Example. A small winery with a few hectares of a rare grape (resource), in a crowded wine market (industry), chooses a limited, premium product rather than volume.

Three value propositions

Customer intimacy

"Choose us because we can customise our products and services to your individual needs better than our competitors".

Think of: A tailor, a private bank, a luxury car built to order.

Operational excellence

"Choose us because we deliver faster, more conveniently and at a lower price than our competitors".

Think of: A discount airline or a low cost retailer.

Product leadership

"Choose us because we offer more innovative and higher quality products than our competitors".

Think of: A pharmaceutical innovator or a technology leader.

Source: Treacy and Wiersema, "Customer Intimacy and Other Value Disciplines", Harvard Business Review, 1993, vol. 71, no. 1

Your turn

Match each company to a proposition

A A sportswear brand launches a new shoe technology every season and patents its materials.

B A supermarket chain keeps a narrow range, simple stores and the lowest shelf prices in town.

C A drug company wants to be “first to market and first in quality” with new compounds.

D A hotel group stores each guest's preferences so they never repeat them in any of its resorts.

E An online retailer promises next day delivery on millions of items and easy returns.

F A luxury car maker offers so many leather, colour and wood options that every car is almost unique.

Solution

Look for the key word

	Proposition	The clue
A	Product leadership	New technology every season, patents
B	Operational excellence	Narrow range, simple stores, lowest price
C	Product leadership	First to market, first in quality
D	Customer intimacy	Individual preferences remembered everywhere
E	Operational excellence	Speed and convenience at scale
F	Customer intimacy	Every car tailored to one buyer

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The framework

Section II

The Balanced Scorecard

Strategy on the left, measures on the right

Vision and strategy

What are our financial goals?

Which customers do we want to serve, and how will we win and keep them?

Which internal business processes are critical to delivering value to customers?



The financial perspective

- **Guiding question:** has our financial performance improved?
- This is what shareholders ultimately care about. It sits at the top because all other improvements should lead to better results here.
- If it never does, **the strategy itself may be wrong.**

Typical measures

Sales growth rate ↑

Net operating income ↑

Residual income ↑

Return on assets ↑

Net cash flow from operations ↑

Earnings per share ↑

The customer perspective

- **Guiding question:** do customers recognise that we are delivering more value?
- The key word is **recognise**. It is not enough that we improve: the customer must notice it and change behaviour, buy again, buy more, recommend us.
- These measures connect what we do inside to what we earn outside.

Typical measures

Market share ↑

Customer defection rate ↓

Number of customer referrals ↑

Share of wallet ↑

Customers who say “my order was delivered on time” ↑

The internal business process perspective

- **Guiding question:** have we improved key business processes so that we can deliver more value to customers?
- Internal business processes are **what the company does** to satisfy customers: assembling a product, handling baggage at an airline.
- Here we measure how well, how fast and how cheaply those processes run.

Typical measures

New products designed ↑

Warranty claims ↓

Cost of quality ↓

Delivery cycle time ↓

Manufacturing cycle efficiency ↑

Overall equipment effectiveness ↑

The learning and growth perspective

- **Guiding question:** are we maintaining our ability to change and improve?
- This is the foundation. Processes are improved by people, so we ask whether employees are hired well, trained, motivated, healthy and willing to stay.
- These measures move slowly, but everything above depends on them.

Typical measures

Training hours per employee ↑

Employee turnover ↓

Absenteeism rate ↓

Improvement suggestions per employee ↑

Leadership roles filled internally ↑

The strategy decides what goes in each box

Example: two pizzerias in the same street. **A** wants to be the quickest and cheapest, **B** the best pizza in town. Read each column from the bottom up.

Perspective	Pizzeria A: fast and cheap	Pizzeria B: the best in town
Financial	Cost per pizza ↓, pizzas sold per evening ↑	Average price per pizza ↑
Customer	Repeat orders per customer ↑	Online reviews that praise the taste ↑
Internal processes	Minutes from order to pizza ↓	New seasonal pizzas on the menu ↑
Learning and growth	Staff trained to work at every station ↑	Pizza makers trained on dough and toppings ↑

Same street, same oven, different strategies: **different measures in every box**. A scorecard copied from a competitor measures the wrong things.

The four perspectives form a chain



Read it from the bottom.

Employees need to keep learning in order to improve internal business processes.

Improving processes is necessary to improve customer satisfaction.

Improving customer satisfaction is necessary to improve financial results.

Every arrow is a hypothesis we can test

If we improve this measure, then that other measure should also improve.

Example

If average training hours per employee go up, then the number of errors per order should go down.

Test

After some months, did errors actually fall? If not, either the training is poor or errors have a different cause.

A strategy map puts the chain on one page

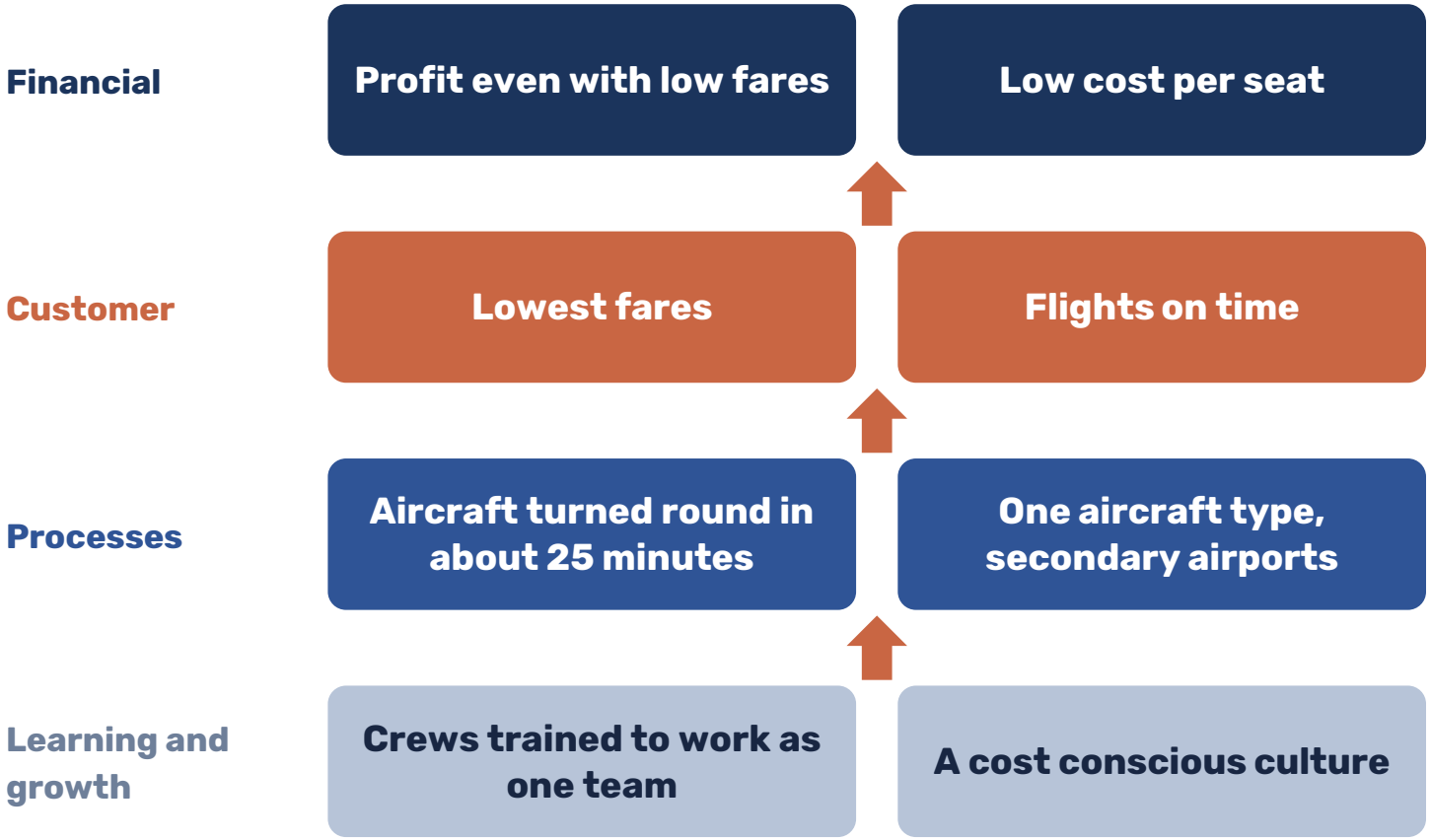
A **strategy map** is a diagram of the strategy:

- the **objectives** in each of the four perspectives
- **arrows** showing which objective should cause which
- everything on **one page**, so each employee sees how their work leads to the results

Then each objective gets a **measure**, a **target** and an **initiative**: that is the scorecard.



Ryanair's strategy map



Read it from the bottom.

- **Fast turnarounds** let each aircraft fly more trips a day, so fewer aircraft serve the same routes.
- Fewer aircraft, all of one type, keep the **cost per seat** low.
- If turnarounds slipped to 45 minutes, **the whole map would weaken**.

From the map to the scorecard

Perspective	Objective	Measure	Target	Initiative
Financial	Low cost per seat	Cost per passenger ↓	Lowest in Europe	Large orders of a single aircraft type
Customer	Flights on time	Flights on time ↑	90%	Daily punctuality report for each base
Processes	Fast turnaround	Minutes from landing to take off ↓	25	Front and rear stairs, no assigned gates
Learning and growth	Crews work as one team	Crew trained on the turnaround routine ↑	100%	Training for every new crew member

Map = **which objectives cause which**. Scorecard = **how we measure each objective, and how far we want to go**. Targets shown are illustrative.

Two simple rules for every measure

RULE 1

Give it an arrow

“Employee turnover: 12%”. Good or bad? On its own we cannot tell.

“Employee turnover ↓”. Now everybody knows the goal: bring it down.

Without an arrow a number is just information. **With an arrow it becomes a goal.**

Market share ↑ Delivery time ↓ Defects ↓ Training hours ↑

RULE 2

A euro sign does not make it financial

Ask what the measure is **about**, not what unit it

Measure in euro	Perspective
Cost of quality	Internal process
Training spending per employee	Learning and growth
Customer lifetime value	Customer
Net operating income	Financial

Three common misreadings

What people sometimes think	What the scorecard really is
"It is a long list of KPIs".	A short set of measures chosen because they follow from the strategy and are linked to each other.
"Improve the operational numbers and profit will follow".	The financial perspective exists to check whether that really happens. If it does not, the strategy must be reconsidered.
"The same scorecard works for every company".	Measures differ from company to company because strategies differ.



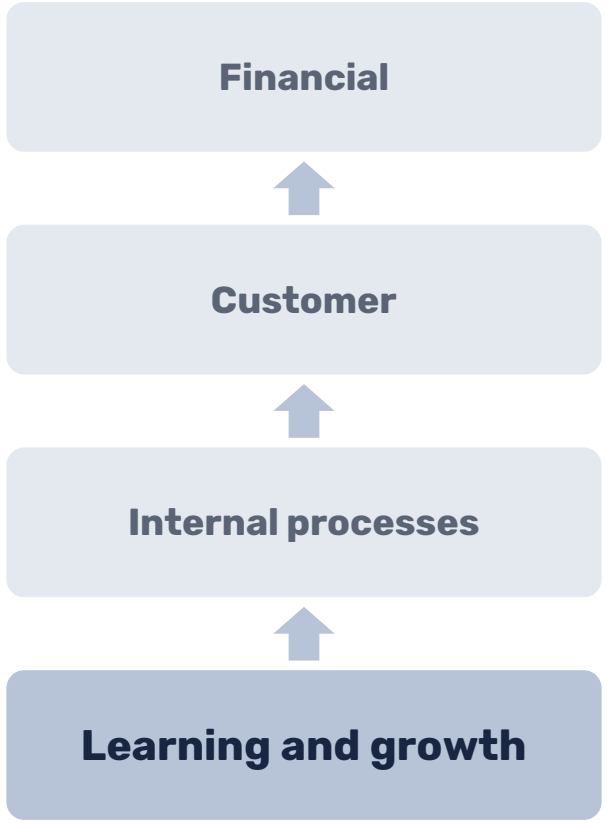
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People, processes and customers

Section III

The Balanced Scorecard

The learning and growth perspective



GUIDING QUESTION

Are we maintaining our ability to change and improve?

What we measure

The skills, motivation, health and stability of the people who run the processes. It is the foundation: everything above depends on it.

Typical measures

Training hours per employee ↑

Employee turnover ↓

Absenteeism rate ↓

Improvement suggestions per employee ↑

Leadership roles filled internally ↑

Next level up: better trained and more stable people should run **better processes**. These measures move slowly.

“Our employees are our most important asset”

- Business leaders say this often. The balanced scorecard asks them to take it seriously and **measure** it.
- The reasoning: if a company with a **viable strategy** lets its employees keep learning and expanding their capabilities, financial performance should ultimately improve.

Note the condition

“With a viable strategy”. Brilliant, well trained people cannot save a strategy that customers do not want.

Learning and growth is **the base of the chain**, not a substitute for the chain.

Six areas that drive employee learning and growth

Recruiting

Do we attract the best candidates at a sensible cost?

Skills development

Do people get the training, mentoring and resources they need?

Pay and advancement

Can good people grow, earn more and be promoted from within?

Wellness and safety

Are people healthy, present and safe at work?

Job satisfaction

Do people like their work and feel their contribution is recognised?

Retention

Do good people stay long enough to build experience?

Learning and growth

Recruiting, skills, pay and advancement

Recruiting

- ↑ Percent of "outstanding" candidates who accepted our job offer
- ↓ Recruiting expenditure per hired employee

Skills

- ↑ Percent of employees who strongly agree "I am consistently given the resources I need to do my job"
- ↑ Average mentorship hours per employee
- ↑ Average training hours per employee

Advancement

- ↑ Percent of leadership positions filled with internal candidates
- ↑ Percent of workforce promoted in the last three years

Pay

- ↑ Average pay raise percentage per employee

↑ goal: increase ↓ goal: decrease

Learning and growth

Wellness, satisfaction and retention

Wellness

↓ Absenteeism rate

Safety

↓ Average number of workplace accidents per employee

Satisfaction

↑ Percent who strongly agree "this is a great place to work"

↑ Percent who strongly agree "my contribution is recognised and respected by my co-workers"

↑ Number of process improvement suggestions per employee

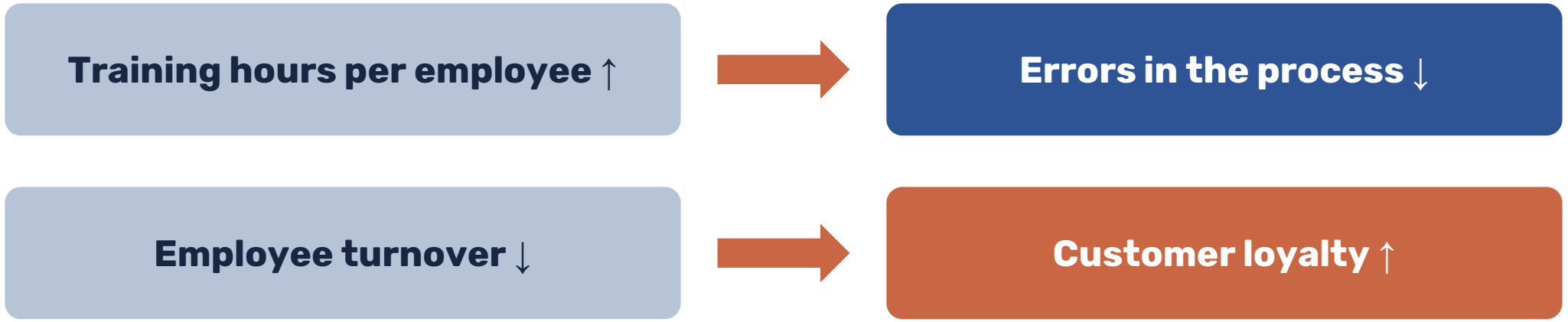
Retention

↓ Employee turnover percentage

↑ Average years of tenure per employee

↑ goal: increase ↓ goal: decrease

A people measure earns its place only if it moves something else



The second link is the “familiar face” argument: customers like dealing with the same people, so lower turnover can build loyalty. **Both links are hypotheses, not facts.**

Employee wellness pays: an Italian example

Stress and illness raise absenteeism and lower productivity. Many companies now treat employee wellness as an investment.

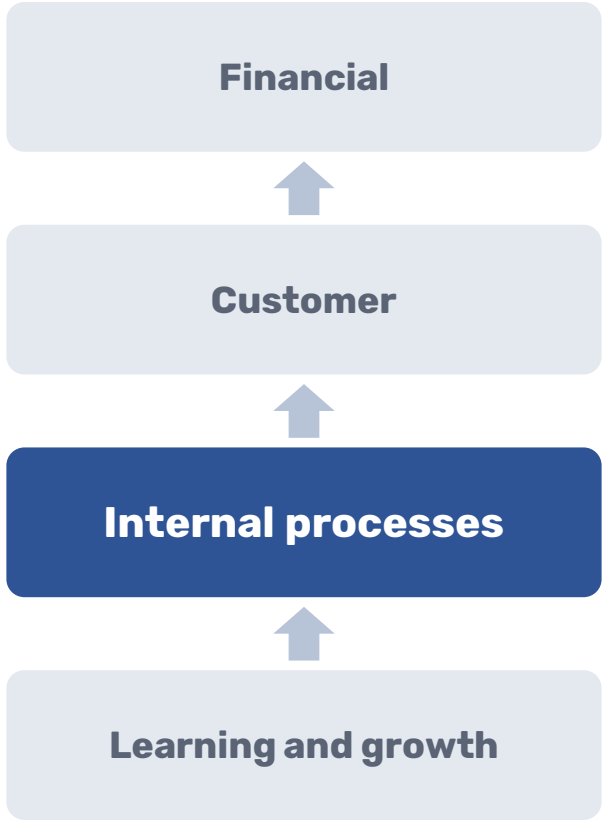
AGORDO, BELLUNO

Luxottica: welfare as part of the strategy

- One of the first Italian companies to build a company welfare system: supplementary healthcare, free checkups for employees over 50, scholarships for employees' children.
- Agreement for 2024 to 2026: a **four day week for 20 weeks a year** at six Italian plants, with the same pay. It covers about 15,000 employees and is voluntary.
- **Scorecard measures it should move: absenteeism ↓, turnover ↓, applications per vacancy ↑.**

Source: Il Giornale, 30 November 2023 (Luxottica)

The internal process perspective



GUIDING QUESTION

Have we improved key processes so that we deliver more value to customers?

What we measure

How well, how fast and how cheaply the company runs the activities that serve customers, such as assembling a product or handling baggage.

Typical measures

New products designed ↑

Warranty claims ↓

Cost of quality ↓

Delivery cycle time ↓

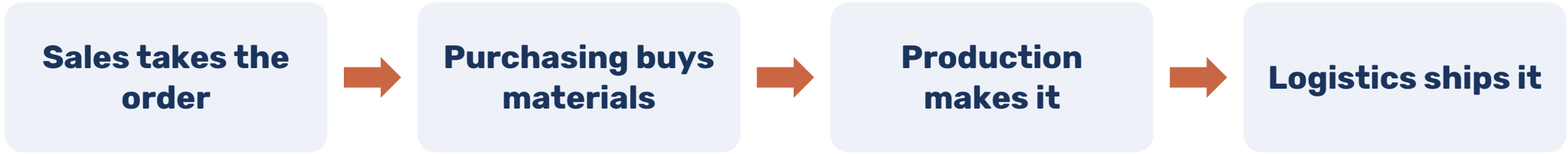
Manufacturing cycle efficiency ↑

Overall equipment effectiveness ↑

Next level up: better processes should give **customers** something they notice and value.

Processes cross departments

A **business process** is a series of steps followed to carry out a task. Customers are served by processes, not by departments.



The customer only sees how long the whole order took and whether it arrived right. Improving a process measure therefore forces managers of different departments to **cooperate**.

Six dimensions of process performance

Innovation

How many new products and patents do we create?

Quality

How often do our products and services work right the first time?

Agility

How quickly can we adapt or change?

Cost

How much do we spend on things that add no value?

Time

How long from order to shipment, and how much of it is real work?

Lean resources

Are equipment, inventory and workspace used without waste?

Internal processes

Innovation, quality and agility

Innovation

- ↑ Number of new products designed
- ↑ Number of patents approved
- ↑ Percentage of previously designed parts used in new products

Quality

- ↓ Number of warranty claims
- ↑ Defect free units as a percentage of completed units
- ↑ Percent of customer complaints settled on first contact

Agility

- ↑ Number of modular product designs
- ↑ Percent of suppliers with long term contracts
- ↓ Average batch size

Product leaders look at the **innovation** column first; customer intimate companies at **agility**.

↑ goal: increase ↓ goal: decrease

Internal processes

Cost, time and lean resource management

Cost

- ↓ Cost of quality
- ↓ Non value added activity costs
- ↓ Lost sales due to out of stock merchandise

Time

- ↓ Delivery cycle time
- ↓ Throughput, or manufacturing cycle, time
- ↑ Manufacturing cycle efficiency (MCE)

Lean

- ↑ Overall equipment effectiveness (OEE)
- ↓ Work in process inventory as a percent of sales
- ↑ Workstation cleanliness score

Operational excellence lives here: **lower cost, shorter times, less waste**.

↑ goal: increase ↓ goal: decrease

Each strategy looks at its processes with one question

PRODUCT LEADERSHIP

“Is it new and better?”

Watch innovation and quality:

- New products designed ↑
- Patents approved ↑
- Defects ↓

Think of Apple

OPERATIONAL EXCELLENCE

“Is it cheap and fast?”

Watch cost and time:

- Cost per unit ↓
- Delivery cycle time ↓
- Waste ↓

Think of Ryanair

CUSTOMER INTIMACY

“Is it made for me?”

Watch service and flexibility:

- Complaints solved at first contact ↑
- Custom orders on time ↑
- Small batch sizes ↓

Think of Ritz Carlton

Every company watches all three, but the strategy decides **which one comes first** on a short scorecard.

Stellantis builds 14 brands on a few shared platforms

- Stellantis was born in 2021 from the merger of FCA and PSA. Its 14 brands range from Fiat and Alfa Romeo to Peugeot, Jeep and Maserati.
- In July 2021 it announced that its electric cars would be built on **four shared platforms**, STLA Small, Medium, Large and Frame, with three common electric drive modules.
- Same idea as Toyota and Volkswagen before it: share components across models to buy in bulk, design faster and spend less on plants.
- The risk: cars of different brands that feel too similar.

QUESTION FOR YOU

Which internal process measure would help Stellantis reach its goal?

Possible answers:

- Share of models built on the four platforms ↑
- Share of parts common across brands ↑
- Months from design to launch ↓

Your turn

Which strategy would adopt each measure?

For each internal process measure choose **operational excellence**, **product leadership** or **customer intimacy**, and give a one line reason.

1. Delivery cycle time ↓
2. Number of new products designed ↑
3. Average contact hours per customer ↑
4. Non value added activity costs ↓
5. Suppliers with long term contracts ↑
6. Number of modular product designs ↑
7. Customised products co designed with customers ↑
8. Defect free units as a percent of completed units ↑
9. Number of options available to customers ↑

Example: patents approved ↑ is product leadership, because patents are the base for innovative products.

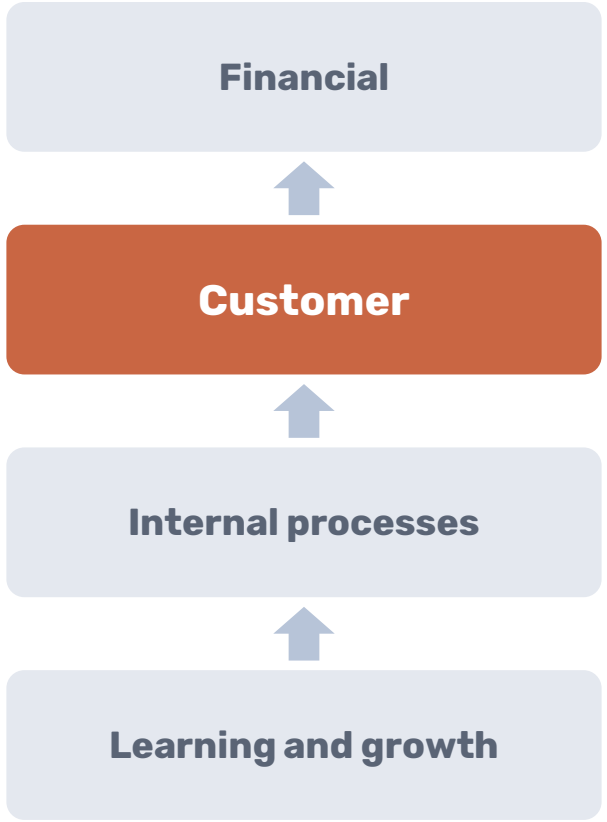
Solution

Measures and the strategies behind them

	Measure	Strategy	Reason
1	Delivery cycle time ↓	Operational excellence	Speed and convenience
2	New products designed ↑	Product leadership	A steady flow of innovation
3	Contact hours per customer ↑	Customer intimacy	Knowing each client's needs
4	Non value added costs ↓	Operational excellence	Lower cost allows lower prices
5	Long term supplier contracts ↑	Customer intimacy *	Partners able to meet special requests
6	Modular product designs ↑	Customer intimacy	Modules make tailoring cheaper
7	Products co designed with clients ↑	Customer intimacy	Built around one customer
8	Defect free units ↑	Product leadership	Superior product quality
9	Options available ↑	Customer intimacy	More ways to tailor the product

** Item 5 can also be argued as operational excellence: stable suppliers mean reliable supply and lower purchasing cost.*

The customer perspective



GUIDING QUESTION

Do customers recognise that we are delivering more value?

What we measure

Whether customers notice our improvements and change behaviour: they buy again, buy more and recommend us. The key word is recognise.

Typical measures

Market share ↑

Customer defection rate ↓

Customer referrals ↑

Share of wallet ↑

Orders delivered on time, as customers see it ↑

Next level up: customers who buy more and stay longer should produce **better financial results**.

Inside improvements must reach the customer

- If a company improves its learning and growth and its process measures, those gains must **ultimately** show up in customer measures.
- If they do not, **the company does not fully understand what shapes its customers' perceptions and behaviour.**

Example of customer measures

- Customer satisfaction
- Acquisition and retention
- Customer lifetime value
- Customer loyalty
- Customer service
- Customer value proposition

Customer

Satisfaction, acquisition and lifetime value

Satisfaction

- ↑ Percent who strongly agree "I would readily recommend your company"
- ↑ Number of customer referrals

Acquisition

- ↑ Market share percentage
- ↓ Average customer acquisition cost
- ↑ Percent of customer leads that generate a sale

Retention

- ↓ Customer defection rate

Lifetime value

- ↑ Average revenue per order
- ↑ Average gross margin per order
- ↓ Average time between purchases

↑ goal: increase ↓ goal: decrease

Customer

Loyalty, service and value proposition

Loyalty

- ↑ Share of the customer's spending committed to us ("share of wallet")
- ↑ Number of collaborative, customer driven product innovations

Service

- ↑ Percent who strongly agree "my order was delivered on time"
- ↑ Percent who strongly agree "my problem was resolved quickly"
- ↑ Percent who strongly agree "your website was easy to use"
- ↑ Percent who strongly agree "your employees treated me courteously"

Proposition

- ↑ Customer perception of our product leadership
- ↑ Customer perception of our operational excellence
- ↑ Customer perception of our customer intimacy

↑ goal: increase ↓ goal: decrease



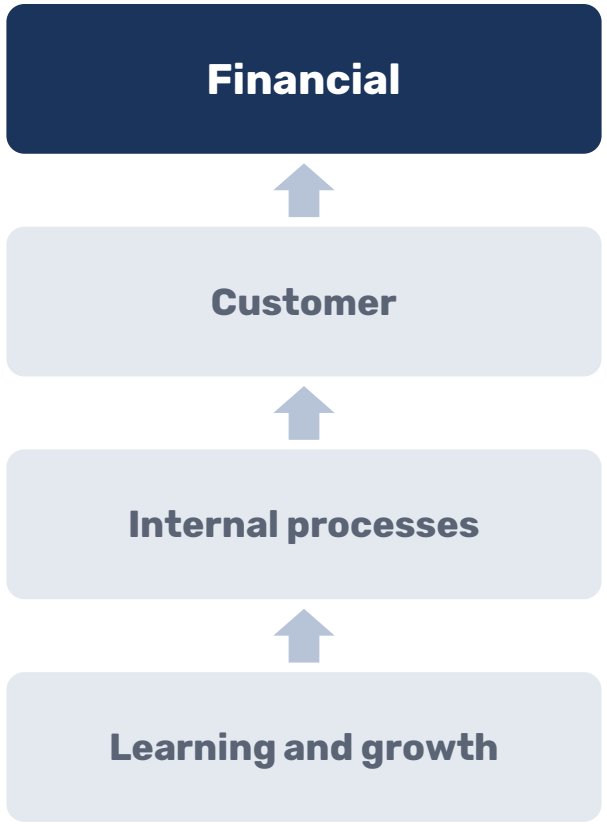
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Financial measures

Section IV

The Balanced Scorecard

The financial perspective



GUIDING QUESTION

Has our financial performance improved?

What we measure

The results shareholders care about. It is the final check: if better people, processes and customers never show up here, the strategy itself may be wrong.

Typical measures

Sales growth rate ↑

Net operating income ↑

Residual income ↑

Return on assets ↑

Net cash flow from operations ↑

Earnings per share ↑

In this section: where these numbers come from, and how ratios turn the statements into measures.

Financial

Sales, profits and profitability ratios

Sales

- ↑ Sales from new customers
- ↑ Sales from products less than three years old
- ↑ Patent protected sales

Profits

- ↑ Contribution margin
- ↑ Net operating income
- ↑ Residual income

Ratios

- ↑ Gross margin percentage
- ↑ Net profit margin percentage
- ↑ Return on assets (ROA)
- ↑ Return on equity (ROE)

Sales measures can carry the strategy: **sales from new products** fit product leadership.

↑ goal: increase ↓ goal: decrease

Financial

Trends, cash flows and market performance

Trend

- ↑ Sales growth rate
- ↑ Contribution margin growth rate
- ↑ Net operating income growth rate
- ↑ Operating cash flow growth rate

Cash flows

- ↑ Net cash flow from operating activities
- ↑ Operating cash flow ÷ average operating assets
- ↑ Operating cash flow ÷ sales

Market

- ↑ Stock price
- ↑ Earnings per share (EPS)
- ↑ Price earnings ratio

Cash flows and market measures show what profit alone cannot: **liquidity** and **investors' expectations**.

↑ goal: increase ↓ goal: decrease

The scorecard does not assume that better processes automatically bring better profits. It checks.

The financial perspective holds the organisation accountable for turning non-financial improvements into bottom line results.

If favourable trends in learning, processes and customers do not translate into financial results, the scorecard forces the company to look again at how it differentiates itself from competitors.

Where the financial measures come from

Many measures in the financial perspective of the scorecard are **ratios** calculated in financial statement analysis, which managers use to assess the financial health of their companies.

Examine trends

How key financial data change over time.

Compare companies

How our figures compare with competitors or the industry.

Analyse ratios

Relations between items that reveal profitability and market value.

Example: **Norton Corporation**, a textbook company with two years of statements.

Basics: the income statement

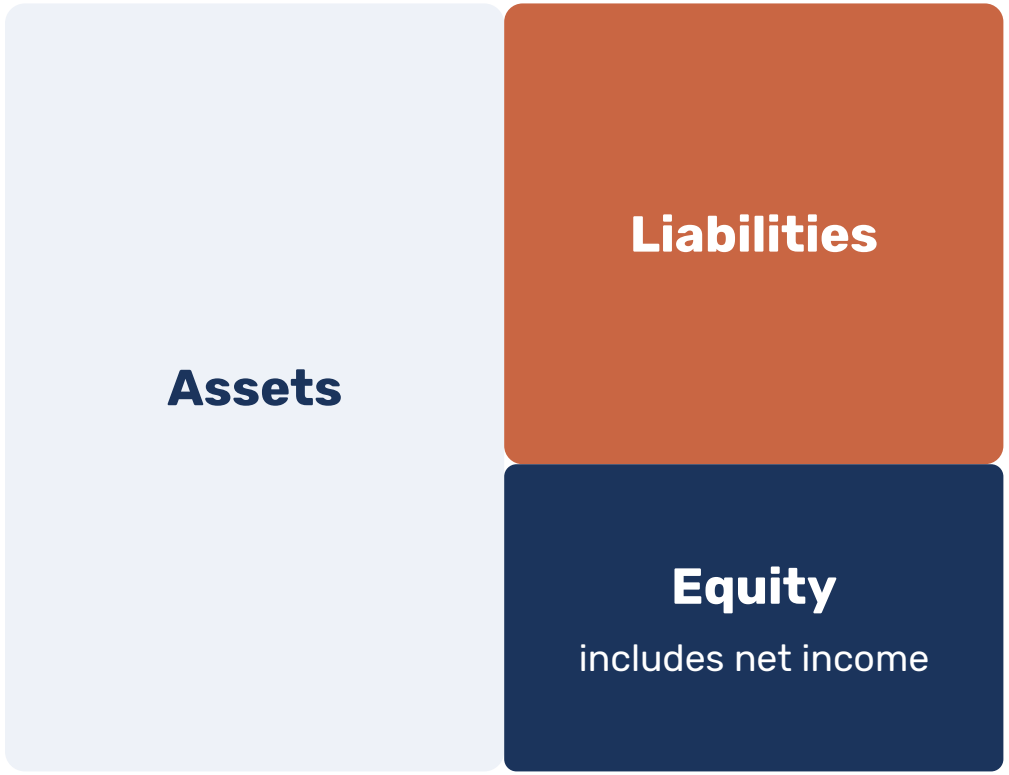
- The financial statement is a **mandatory** set of documents prepared yearly by firms.
- It includes two main accounting documents: the income statement and the balance sheet.
- The **income statement** compares positive and negative items of income over the year.



Net income realised during the financial year.

Basics: the balance sheet

- The **balance sheet** shows, at the end of the financial year, what the company owns and how it is financed.
- Assets = Liabilities + Equity.
- The net income of the year becomes part of equity.
- To assess financial health, the items of both statements are usually rearranged ("analysed") to obtain more information.



Assets

Item	Meaning
Cash and cash equivalents	Money held and amounts in checking and savings accounts
Accounts receivable	Amounts due from customers for goods or services already delivered or sold
Inventories	Goods kept in stock, available for sale
Property, plant and equipment	Assets that cannot readily be converted into cash: computers, cars, plants
Securities	Bonds, stocks or shares of other firms acquired for investment purposes

Liabilities

Item	Meaning
Bank overdraft	Short term borrowing from the bank on the current account, often used to pay suppliers
Accounts payable	Amounts due to suppliers for goods or services already received
Notes payable	Obligation to repay borrowed money at a particular future date
Severance provisions	Obligations toward employees, paid when their employment ends
Long term borrowings	Obligations to lenders, such as bank loans, due beyond 12 months

Equity

Item	Meaning
Share capital	Amounts invested by the owners of the company
Par value	The nominal value of each share
Additional paid in capital	Money investors pay above the par value of the shares
Retained earnings	Net income accumulated over the life of the company and not distributed as dividends
Net income	Difference between positive and negative items of income: profit or loss

Norton Corporation: balance sheet, assets

Assets	This year	Last year
Cash	$30,000	$20,000
Accounts receivable, net	20,000	17,000
Inventory	12,000	10,000
Prepaid expenses	3,000	2,000
Total current assets	65,000	49,000
Land	165,000	123,000
Buildings and equipment, net	116,390	128,000
Total property and equipment	281,390	251,000
Total assets	**$346,390**	**$300,000**

Balance sheets at 31 December.

Total assets grew by $46,390.

Norton Corporation: liabilities and equity

Liabilities and equity	This year	Last year
Accounts payable	$39,000	$40,000
Notes payable, short term	3,000	2,000
Total current liabilities	42,000	42,000
Notes payable, long term	70,000	78,000
Total liabilities	112,000	120,000
Common stock, $1 par value	27,400	17,000
Additional paid in capital	158,100	113,000
Retained earnings	48,890	50,000
Total stockholders' equity	234,390	180,000
Total liabilities and equity	**$346,390**	**$300,000**

Par value = nominal value of a share.

Additional paid in capital: money paid above par value.

Norton Corporation: income statement

Income statement	This year	Last year
Sales	$494,000	$450,000
Cost of goods sold	140,000	127,000
Gross margin	354,000	323,000
Operating expenses	270,000	249,000
Net operating income	84,000	74,000
Interest expense	7,300	8,000
Net income before taxes	76,700	66,000
Income taxes (30%)	23,010	19,800
Net income	**$53,690**	**$46,200**

Gross margin = sales – cost of goods sold (product costs only, not selling and administrative).

Interest expense = the cost of financial resources.

A number on its own says little

An item on a financial statement has little meaning by itself. Its meaning grows when we **compare** it, using three tools:

1

Euro and percentage changes

Horizontal or trend analysis: the same item over time.

2

Common size statements

Vertical analysis: each item as a percentage of a total.

3

Ratios

Relations between items, such as margins, returns and market measures.

1. Horizontal analysis: how did each line change?

Horizontal (trend) analysis puts the same line of two or more years side by side and asks: **how much did it change?**

Change = This year − Last year

% change = Change ÷ Last year

EXAMPLE: NORTON'S SALES

This year $494,000, last year $450,000

Change = $494,000 − $450,000 = **$44,000**

% change = $44,000 ÷ $450,000 = **9.8%**

Euro changes

Show which changes **matter most**. An increase of €44,000 in sales weighs more than one of €2,000 in office supplies.

Percentage changes

Show which changes are **unusual**. Office supplies from €2,000 to €4,000 is +100%: small money, but worth a question.

Norton's income statement over two years

$	This year	Last year	Change	Change %
Sales	494,000	450,000	44,000	9.8%
Cost of goods sold	140,000	127,000	13,000	10.2%
Gross margin	354,000	323,000	31,000	9.6%
Operating expenses	270,000	249,000	21,000	8.4%
Net operating income	84,000	74,000	10,000	13.5%
Interest expense	7,300	8,000	(700)	(8.8%)
Net income before taxes	76,700	66,000	10,700	16.2%
Income taxes (30%)	23,010	19,800	3,210	16.2%
Net income	**53,690**	**46,200**	**7,490**	**16.2%**

Sales grew 9.8%, net income 16.2%: costs below gross margin grew more slowly than sales.

2. Common size statements

- **Vertical analysis** focuses on the relations among financial statement items **at a given point in time**.
- A **common size** statement expresses each item as a percentage of a total.
- It makes companies of different size, or years with different volumes, comparable.

Balance sheet

Each item as a percentage of **total assets**.

Income statement

Each item as a percentage of **sales**.

Norton's common size income statement

% of sales	This year	Last year
Sales	100.0%	100.0%
Cost of goods sold	28.3%	28.2%
Gross margin	71.7%	71.8%
Operating expenses	54.7%	55.3%
Net operating income	17.0%	16.4%
Interest expense	1.5%	1.8%
Net income before taxes	15.5%	14.7%
Income taxes	4.7%	4.4%
Net income	**10.9%**	**10.3%**

What it shows

Gross margin is stable: about 72 cents of every sales dollar remain after product costs.

Operating expenses fell from 55.3% to 54.7% of sales.

So net operating income rose from 16.4% to 17.0%, and net income from 10.3% to **10.9%** of sales.

3. Ratios in the financial perspective

Managers compute many ratios for management purposes. In the balanced scorecard they appear under three headings:

Profitability ratios

- Gross margin percentage
- Net profit margin percentage
- Return on assets (ROA)
- Return on equity (ROE)

Profits

- Contribution margin ↑
- Net operating income ↑
- Residual income ↑

Market performance

- Stock price ↑
- Earnings per share (EPS) ↑
- Price earnings ratio ↑

Gross margin percentage

Gross margin percentage = Gross margin ÷ Sales

NORTON CORPORATION, THIS YEAR

= ($494,000 — $140,000) ÷ $494,000

= $354,000 ÷ $494,000

71.7%

What it tells you

How much of each euro of sales is left after paying for the product itself. A high value means pricing power or cheap production; compare it with competitors in the same industry.

In real life

Luxury brands such as Ferrari or Prada keep a large share of each euro of sales. A discount supermarket keeps a small share and must win on volume.

Net profit margin percentage

Net profit margin percentage = Net income ÷ Sales

NORTON CORPORATION, THIS YEAR

= $53,690 ÷ $494,000

10.9%

What it tells you

How much of each euro of sales ends up as profit after all costs, including selling and administrative expenses, interest and taxes.

In real life

Supermarkets often keep only a few cents of profit per euro of sales; large software companies can keep twenty cents or more.

Return on total assets (ROA)

ROA = [Net income + Interest expense × (1 − tax rate)] ÷ Average total assets

NORTON CORPORATION, THIS YEAR

= [$53,690 + $7,300 × 0.70] ÷ [($300,000 + $346,390) ÷ 2]

= $58,800 ÷ $323,195

18.19%

What it tells you

How much profit the company earns on everything it owns. Interest is added back, so that companies with more or less debt can be compared fairly.

In real life

Asset light businesses, such as software or online booking platforms, can reach a high ROA. Railways, steel mills and utilities need huge assets, so their ROA is usually lower.

Return on equity (ROE)

ROE = Net income ÷ Average stockholders' equity

NORTON CORPORATION, THIS YEAR

= $53,690 ÷ [($180,000 + $234,390) ÷ 2]

= $53,690 ÷ $207,195

25.91%

What it tells you

How much profit the company earns for each euro the owners have invested. Shareholders compare it with what they could earn elsewhere.

In real life

Banks such as Intesa Sanpaolo and UniCredit put a return on equity measure at the centre of their results. ROE above ROA, as at Norton, signals the use of debt.

Return on investment (ROI)

ROI = Net operating income (EBIT) ÷ Average operating assets = Margin × Turnover

What it tells you

Whether a division earns enough on the assets it controls. Companies use it to evaluate divisions and their managers.

Operating assets

Cash, receivables, inventory, plant and equipment used to operate, measured at **net book value**.

Margin and turnover

Margin = NOI ÷ sales; turnover = sales ÷ average operating assets.

NET BOOK VALUE

What an asset cost, less the depreciation accumulated so far. A machine bought for €100,000 and depreciated €20,000 a year has, after three years, a net book value of €100,000 − €60,000 = **€40,000**.

Norton (all assets treated as operating): margin = 84,000 ÷ 494,000 = 17.0%; turnover = 494,000 ÷ 323,195 = 1.53; ROI = 17.0% × 1.53 = **26.0%**.

Residual income

Residual income = Net operating income − (Average operating assets × Minimum required rate of return)

What it tells you

Whether the division earns more than the minimum return the owners require. It is an amount in euro, not a percentage.

Why use it besides ROI

A division with an ROI of 25% may reject a project earning 20% because it lowers its average, even if the company needs only 15%. Residual income rewards every project above 15%.

Norton, as an illustration with a 15% minimum rate: 84,000 − (323,195 × 15%) = 84,000 − 48,479 = **$35,521**.

In real life the same idea is known as economic value added (EVA): profit less a charge for all the capital used. Coca Cola made it famous in the 1990s.

Market performance: the data

Norton Corporation, this year	Value
Common shares outstanding, beginning of year	17,000
Common shares outstanding, end of year	27,400
Stockholders' equity	$234,390
Dividends per share	$2
Market price per share, 31 December	$20

Why: net income is earned during the whole year, so we compare it with the resources available on average during that year, not on its last day.

RULE

Film ÷ photo: use the average photo

The **income statement is a film**: it covers the whole year.

The **balance sheet is a photo**: it shows one day.

When a ratio divides a film by a photo, use the **average** of the photos at the start and at the end of the year.

Example: shares for EPS = (17,000 + 27,400) ÷ 2 = 22,200

Earnings per share (EPS)

EPS = Net income ÷ Average number of common shares outstanding

NORTON CORPORATION, THIS YEAR

= $53,690 ÷ [(17,000 + 27,400) ÷ 2]

= $53,690 ÷ 22,200

$2.42

What it tells you

How much of the year's profit belongs to each share. It is the starting point for dividends and for the share price.

In real life

Listed companies announce EPS every quarter, and the press reports whether it beat or missed analysts' forecasts. A miss can move the share price sharply in a single day.

Price earnings ratio

Price earnings ratio = Market price per share ÷ Earnings per share

NORTON CORPORATION, THIS YEAR

= $20.00 ÷ $2.42

8.26×

Investors pay 8.26 dollars for each dollar of current annual earnings.

What it tells you

How many euros investors pay today for one euro of this year's profit. It measures their expectations about future growth, not current performance.

In real life

Fast growing technology companies often trade at high multiples; mature banks and utilities at much lower ones. Norton's 8.26× is a modest multiple.

Norton's financial perspective at a glance

Ratio	Norton, this year	Scorecard heading	Goal
Gross margin percentage	71.7%	Profitability ratios	↑
Net profit margin percentage	10.9%	Profitability ratios	↑
Return on assets	18.19%	Profitability ratios	↑
Return on equity	25.91%	Profitability ratios	↑
Earnings per share	$2.42	Market performance	↑
Price earnings ratio	8.26 times	Market performance	↑

A ratio becomes a scorecard measure only when it has a **direction**, a **target** and a link to the non-financial measures that drive it.

Your turn

Place each measure in its perspective

Learning and growth, internal business process, customer or financial? Example: employee absenteeism rate is learning and growth.

1. Sales from new customers
2. Customer defection rate
3. Average fuel cost per sales dollar
4. Workplace accidents per employee
5. Delivery cycle time
6. Average training hours per employee
7. Job applicants from under represented groups
8. Customers who agree “your employees treated me courteously”
9. Return on assets
10. Customers who agree “your company has a superior commitment to product safety”
11. Number of modular product designs
12. Lost sales due to out of stock merchandise
13. Pounds of waste produced
14. Number of customer referrals
15. Residual income
16. Average mentorship hours per employee

Solution

Four measures in each perspective

Learning and growth	Internal process	Customer	Financial
4. Workplace accidents	5. Delivery cycle time	2. Defection rate	1. Sales from new customers
6. Training hours	11. Modular product designs	8. Courteous treatment	3. Fuel cost per sales dollar
7. Applicants from under represented groups	12. Lost sales from stock outs	10. Product safety commitment	9. Return on assets
16. Mentorship hours	13. Pounds of waste	14. Customer referrals	15. Residual income

Ask what each measure is **about**. Lost sales from stock outs is measured in money, but it describes a failure of the inventory process, so it belongs with internal processes.

Your turn

Label, link and name the strategy

Company A

Average years of tenure per employee · Sales from products less than three years old · Market share percentage · Number of new products designed

Company B

Customer perception of our product leadership · Percent of “outstanding” candidates who accepted our offer · Number of patents approved · Patent protected sales

Company C

Pounds of waste produced · Percent of revenue from eco conscious products · Process improvement suggestions per employee · Customers who agree “your company is committed to environmental stewardship”

Company D

Net operating income growth rate · Manufacturing cycle efficiency · Customers who agree “my order was delivered on time” · Average training hours per employee

For each company: label each measure with perspective and direction, link them with three if/then statements, and name the value proposition.

Solution

Two product leaders

Company A

L&G years of tenure ↑

Process new products designed ↑

Customer market share ↑

Financial sales from products under three years old ↑

If experienced staff stay longer, they design more new products; more new products win market share; more share means more sales from new products.

Company B

L&G outstanding candidates who accept ↑

Process patents approved ↑

Customer perception of product leadership ↑

Financial patent protected sales ↑

If we hire the best people, we obtain more patents; more patents make customers see us as leaders; that perception drives patent protected sales.

Why product leadership: every measure points to new, innovative, protected products.

Solution

One clear case and one to debate

Company C

L&G improvement suggestions ↑
Process pounds of waste ↓
Customer perceived stewardship ↑
Financial revenue from eco products ↑

Open question. Waste reduction and suggestions point to operational excellence; eco products and a green image point to product leadership.

Company D

L&G training hours ↑
Process manufacturing cycle efficiency ↑
Customer “delivered on time” ↑
Financial net operating income growth ↑

Operational excellence. Training makes production more efficient, efficient production delivers on time, reliable delivery grows income.

Company C shows that social and environmental measures fit inside the four perspectives.



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The cost of quality

Section V

The Balanced Scorecard

What are quality costs?

Costs incurred to **prevent** defects, or that **result from** defects in products, are known as quality costs.

Many companies are working hard to reduce them

Quality costs are often large and spread across many departments.

Those that succeed have high quality of conformance

The overwhelming majority of their products conform to design specifications and are free from defects.

What do we mean by quality?

- A product has high **quality of conformance** when it meets or exceeds its design specifications and is free of defects that spoil its appearance or reduce its performance.
- The question is not “is this a luxury product?” but **“does this product do what it was designed to do, every time?”**

An example

A basic city bicycle with brakes that always work has **high** quality of conformance.

An expensive racing bike with a gear that slips does **not**, however advanced its design.

Four types of quality cost

BEFORE DEFECTS

Prevention costs

Support activities whose purpose is to stop defects from occurring at all.

BECAUSE OF DEFECTS

Internal failure costs

Result from defects found **before** the product is shipped: scrap, rework, downtime.

BEFORE DEFECTS

Appraisal costs

Inspection costs: spent to find defective products before they reach customers.

BECAUSE OF DEFECTS

External failure costs

Arise when a defective product reaches the customer: warranty, recalls, lost sales.

Cost of quality = prevention + appraisal + internal failure + external failure. The goal is to lower this total over time.

Prevention and appraisal: examples

Prevention
stop defects from happening

Appraisal
find them before they leave

Internal failure
fix what is found inside

External failure
defects the customer finds

Prevention

- Systems development
- Quality engineering
- Quality training
- Quality circles
- Statistical process control
- Supervision of prevention activities
- Quality data gathering and reporting
- Quality improvement projects
- Technical support to suppliers

Test: does it stop a defect from being created?

Appraisal

- Inspection of incoming materials
- Inspection of work in process
- Final product testing
- Supplies used in testing
- Supervision of testing
- Depreciation of test equipment
- Maintenance of test equipment
- Plant utilities in the inspection area
- Field testing at the customer's site

Test: does it look for defects that may already exist?

Internal and external failure: examples

Prevention
stop defects from happening

→

Appraisal
find them before they leave

→

Internal failure
fix what is found inside

→

External failure
defects the customer finds

Internal failure

- Net cost of scrap
- Net cost of spoilage
- Rework labour and overhead
- Reinspection of reworked products
- Retesting of reworked products
- Downtime caused by quality problems
- Disposal of defective products
- Analysis of the cause of defects
- Correcting data keying errors

Found inside: before the product reaches the customer.

External failure

- Field servicing and handling complaints
- Warranty repairs and replacements
- Repairs beyond the warranty period
- Product recalls
- Liability from defective products
- Returns and allowances from quality problems
- Lost sales from a reputation for poor quality

Found outside: the most expensive place to find a defect.

An example: a brake cable that snaps

Serio Bikes, a bicycle maker, finds that some brake cables snap. The same defect can be handled at four points, and the **cost per defect grows at every step**.

PREVENTION

€2

Agree a stronger cable with the supplier; train assemblers to fit it.

APPRAISAL

€10

Test every brake before the bike is boxed.

INTERNAL FAILURE

€40

Replace the cable in the plant: labour, parts, a delayed order.

EXTERNAL FAILURE

€400 or more

Warranty repair, shipping, an angry customer, bad reviews, and the risk of an accident and a recall.

Costs per defect are illustrative. The further a defect travels, the more it costs.

Spend a little more before, lose much less after

INVEST

Prevention ↑ and appraisal ↑



SAVE

Internal failure ↓ and external failure ↓

- An effective quality programme makes extra investments in prevention and appraisal that are **more than offset** by lower failure costs, so the total cost of quality falls.
- Over time, appraisal should also fall: when defects are prevented at the source, there is less to inspect.

THE RULE TO REMEMBER

Preventing defects is better than detecting them after they have occurred.

Every euro moved from failures to prevention is a sign that the quality system is maturing.

Your turn

Classify nineteen quality activities

Prevention, appraisal, internal failure or external failure? Then say which two types are spent to **keep** poor quality from happening, and which two arise **because** it happened.

a. Product testing

b. Product recalls

c. Rework labour and overhead

d. Quality circles

e. Downtime caused by defects

f. Cost of field servicing

g. Inspection of goods

h. Quality engineering

i. Warranty repairs

j. Statistical process control

k. Net cost of scrap

l. Depreciation of test equipment

m. Returns and allowances from poor quality

n. Disposal of defective products

o. Technical support to suppliers

p. Systems development

q. Warranty replacements

r. Field testing at customer site

s. Product design

Solution

The nineteen activities sorted

Prevention

d. Quality circles

h. Quality engineering

j. Statistical process control

o. Supplier support

p. Systems development

s. Product design

Appraisal

a. Product testing

g. Inspection of goods

l. Depreciation of test equipment

r. Field testing at customer site

Internal failure

c. Rework

e. Downtime from defects

k. Net cost of scrap

n. Disposal of defective products

External failure

b. Product recalls

f. Field servicing

i. Warranty repairs

m. Returns and allowances

q. Warranty replacements

Part 2. Prevention and appraisal are spent to keep poor quality of conformance from reaching customers. Internal and external failure costs are incurred because poor quality has occurred.



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Quality cost reports

Section VI

The Balanced Scorecard

What a quality cost report shows

- A **quality cost report** details prevention costs, appraisal costs and the costs of internal and external failures arising from the company's current quality efforts.
- It is used to monitor progress in lowering the overall cost of quality, usually by comparing two or more periods.

How it is laid out

1. The four categories, each with its detailed lines and a subtotal
2. A grand total: the cost of quality
3. Each amount also shown as a **percentage** of a base, usually total sales
4. One column per year, side by side

Serio Bikes: quality costs for two years

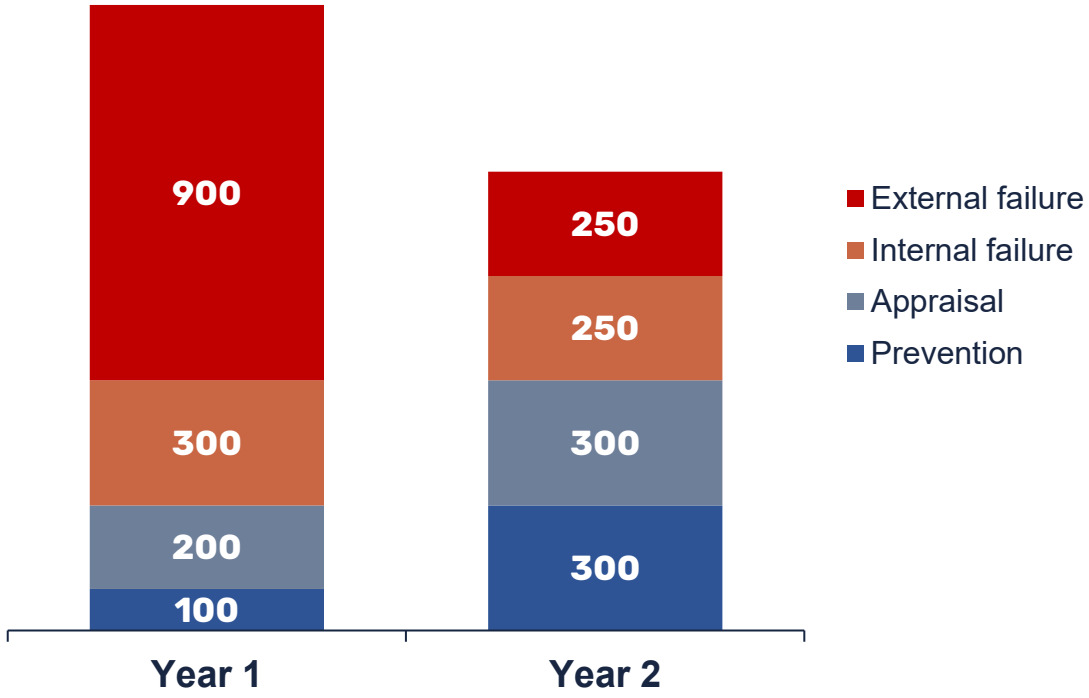
Sales of €**10 million** in each year. **Year 1:** many bikes came back under warranty. **Year 2:** the company trained its welders and checked suppliers before buying.

€ thousands	What it includes	Year 1	% of sales	Year 2	% of sales
Prevention	Training welders, selecting suppliers	100	1.0%	300	3.0%
Appraisal	Testing frames and brakes	200	2.0%	300	3.0%
Internal failure	Reworked and scrapped frames	300	3.0%	250	2.5%
External failure	Warranty repairs, returned bikes	900	9.0%	250	2.5%
Total cost of quality		**1,500**	**15.0%**	**1,100**	**11.0%**

Each percentage is the cost divided by sales. Example: 900 ÷ 10,000 = **9.0%**.

Where Serio's quality money went, € thousands

Total: **1,500** in year 1, **1,100** in year 2



+300

spent **before**: prevention +200, appraisal +100

(700)

saved **after**: internal failure (50), external failure (650)

(400)

net change in the total cost of quality

Two tests the report passes

TEST 1 ✓

More spent before, much more saved after

Prevention and appraisal rose by **300**; failure costs fell by **700**.

Each extra euro spent before saved about €**2.30** after: 700 ÷ 300 = 2.33.

TEST 2 ✓

The total falls as a share of sales

Year 1: 1,500 ÷ 10,000 = **15%**.

Year 2: 1,100 ÷ 10,000 = **11%**.

Four cents of every euro of sales are no longer lost to poor quality.

These are the two trends managers look for in every quality cost report, year after year.

Reading Serio's report: three observations

1

Year 1 was poorly distributed

80% of the quality money went to repairing failures, only 20% to preventing and finding them: 1,200 of 1,500.

2

Year 2 moved the money upstream

Prevention tripled, from 100 to 300, and warranty costs fell from 900 to 250. Fixing the cause was cheaper than paying for the effects.

3

The next step: less inspection

As defects become rare, the company can test less. Appraisal itself should eventually fall.

Why managers find the report useful

It shows the size

Quality costs cut across departments and are not normally collected by the cost system. Managers are often surprised by how large they are.

It ranks problems

It reveals whether the big issue is scrap, warranty costs or something else, so managers know where to focus first.

It shows the mix

It tells managers whether money is spent preventing defects or repairing them. A good mix leans toward prevention and appraisal; a poor one is dominated by failures.

What the report cannot do

It does not fix anything

Measuring and reporting quality costs does not solve quality problems. Only action does.

Results come late

At first the total may even rise while new quality systems are designed and installed. Savings often appear only later.

The biggest cost is missing

Lost sales because unhappy customers stop buying are usually not included because they are hard to estimate. Yet it is often the most important quality cost.

Your turn

Is Walton's cost of quality trending well?

Category	Last year	This year
Prevention	$357,000	$650,000
Appraisal	$445,000	$545,000
Internal failure	$790,000	$500,000
External failure	$1,100,000	$680,000

Compute

1. Total cost of quality for each year
2. Each category as a percent of the total, last year
3. The same percentages for this year
4. Is performance moving in a favourable direction? Why?

Solution

Lower total, better mix

Category	Last year	% of total	This year	% of total
Prevention	$357,000	13.3%	$650,000	27.4%
Appraisal	445,000	16.5%	545,000	22.9%
Internal failure	790,000	29.3%	500,000	21.1%
External failure	1,100,000	40.9%	680,000	28.6%
Total	**$2,692,000**	**100%**	**$2,375,000**	**100%**

Favourable. Walton spent $393,000 more on prevention and appraisal and saved $710,000 on failures, so the total fell by $317,000. Failures dropped from 70.2% to 49.7% of the total. Next goal: keep shifting spending toward prevention.

Your turn

Has Mercury's quality programme worked?

Mercury makes mobile phones in Texas and is losing share to overseas rivals that compete on price and quality. A survey ranked its phones low in quality, so the president launched a company wide improvement programme.

The production manager, John Tran, was sceptical: “Quality is too abstract to be attaching costs to it”.

Task: show each cost as a % of total production cost and of total quality cost. Has the programme succeeded?

$ thousands	Last year	This year
Prevention	70	150
Appraisal	100	130
Internal failure	90	200
External failure	410	110
Total quality cost	**670**	**590**
Total production cost	4,200	4,800

Solution

The expanded report

Category	Last: % prod.	Last: % qual.	This: % prod.	This: % qual.
Prevention	1.7%	10.4%	3.1%	25.4%
Appraisal	2.4%	14.9%	2.7%	22.0%
Internal failure	2.1%	13.4%	4.2%	33.9%
External failure	9.8%	61.2%	2.3%	18.6%
Total	**16.0%**	**100%**	**12.3%**	**100%**

Yes, it is working. Total quality cost fell from $670k to $590k while production cost rose, from 16.0% to 12.3% of production cost. External failure fell from 61.2% to 18.6% of quality costs, with customer returns down from $320k to $80k.

Workload, and the cost of doing nothing

Will production stay this busy?

Tran is right that rework rose from $50k to $130k: defects are now caught in the plant instead of by customers.

As prevention matures, fewer defects are created, so rework, scrap and the workload in production should fall over time.

What if Mercury had done nothing?

The real cost is lost sales and market share from customers who leave because of poor quality.

Mercury could estimate it from the market share trend, survey data on why customers switch, and the margin lost per customer.

Your turn

Build the report from a raw list

$ thousands	Last	This
Inspection	750	900
Quality engineering	420	570
Depreciation of test equipment	210	240
Rework labour	1,050	1,500
Statistical process control	0	180
Cost of field servicing	1,200	900
Supplies used in testing	30	60

$ thousands	Last	This
Systems development	480	750
Warranty repairs	3,600	1,050
Net cost of scrap	630	1,125
Product testing	810	1,200
Product recalls	2,100	750
Disposal of defective products	720	975

Florex installed statistical process control to cut rising warranty and field costs. Sales were flat at **$75,000,000** a year. **1.** Prepare a quality cost report for both years, with percentages of sales to two decimals. **2.** Write a short evaluation for management.

Solution

Florex quality cost report, $ thousands

Category and lines	Last year	% sales	This year	% sales
Prevention: quality engineering, statistical process control, systems development	900	1.20%	1,500	2.00%
Appraisal: inspection, test equipment, supplies, product testing	1,800	2.40%	2,400	3.20%
Internal failure: rework, scrap, disposal	2,400	3.20%	3,600	4.80%
External failure: field servicing, warranty repairs, recalls	6,900	9.20%	2,700	3.60%
Total cost of quality	**12,000**	**16.00%**	**10,200**	**13.60%**

The money is moving to the right place

Distribution

Last year 77.5% of quality costs were failures (9,300 of 12,000) and external failure alone was 9.20% of sales. This year failures are 61.8% (6,300 of 10,200).

Change

Prevention and appraisal each rose by $600k. Internal failure rose by $1.2m as more defects were caught in the plant. External failure fell by $4.2m.

Result

Total cost of quality fell by $1.8m, from 16.00% to 13.60% of sales. Keep investing in prevention so internal failures, and later appraisal, also fall.

Your turn

Company Alfa: quality costs over two years

Alfa manufactures and sells industrial grinders. Figures in € thousands. Sales: **10,000** in year N and **12,500** in year N+1.

€000	Year N	Year N+1
Line inspection	110	85
Scrap	250	175
Design engineering	100	240
Cost of returned goods	60	145
Product testing equipment	50	50
Customer support	40	30
Rework costs	160	135
Preventive equipment maintenance	35	90

€000	Year N	Year N+1
Product liability claims	200	100
Incoming materials inspection	20	40
Breakdown maintenance	90	40
Product testing labour	220	75
Training	45	120
Warranty repair	300	200
Supplier evaluation	20	50

Your turn

Company Alfa: requirements

1. Classify the cost items into the four categories of the cost of quality (COQ).

2. Calculate the ratio of each COQ category to sales in year N and year N+1. Comment on the trend between the two years.

3. Give some examples of non-financial quality measures that Alfa could monitor as part of a total quality control effort.

Solution

Alfa solution: prevention and appraisal

€000	Year N	% sales	Year N+1	% sales
Design engineering	100		240	
Preventive maintenance	35		90	
Training	45		120	
Supplier evaluation	20		50	
Total prevention	**200**	**2.0%**	**500**	**4.0%**
Line inspection	110		85	
Product testing equipment	50		50	
Incoming materials inspection	20		40	
Product testing labour	220		75	
Total appraisal	**400**	**4.0%**	**250**	**2.0%**

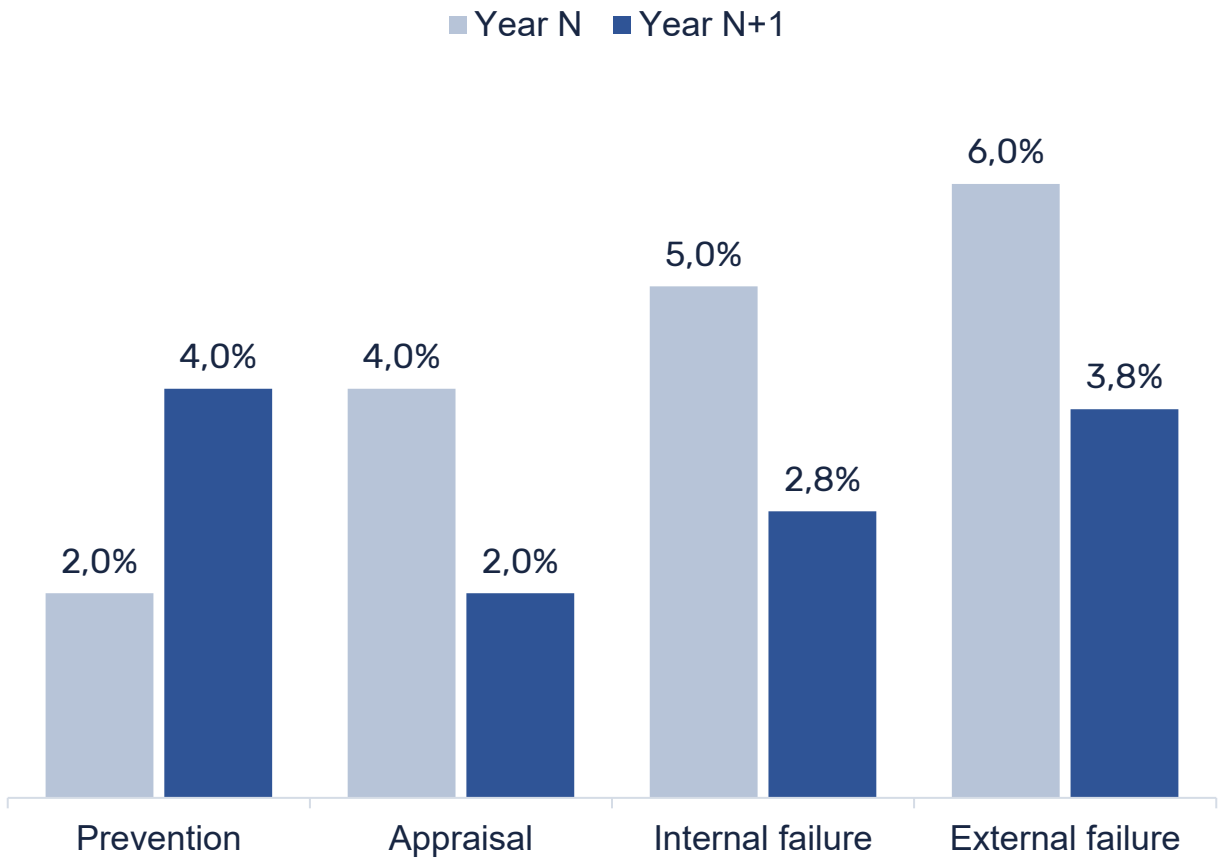
Solution

Alfa solution: failure costs and total

€000	Year N	% sales	Year N+1	% sales
Scrap	250		175	
Rework	160		135	
Breakdown maintenance	90		40	
Total internal failure	**500**	**5.0%**	**350**	**2.8%**
Returned goods	60		145	
Customer support	40		30	
Product liability claims	200		100	
Warranty repair	300		200	
Total external failure	**600**	**6.0%**	**475**	**3.8%**
Total cost of quality	**1,700**	**17.0%**	**1,575**	**12.6%**

Alfa's cost of quality fell from 17% to 12.6% of sales

Solution



- Alfa moved resources **into prevention**: design engineering, preventive maintenance, training and supplier evaluation. Prevention rose from 2% to 4% of sales.
- As a result appraisal fell from 4% to 2%, internal failure from 5% to 2.8%, external failure from 6% to 3.8%.
- Total cost of quality: **17% to 12.6% of sales**.

Solution

Alfa: non-financial quality measures

Examples of non-financial quality measures Alfa could monitor:

CUSTOMER

Defective units shipped

Number of defective products shipped to customers as a % of total units shipped ↓

PROCESS

Good output ratio

Ratio of good output to total output at each production process ↑

LEARNING AND GROWTH

People

Employee training hours ↑ and employee turnover ↓

CUSTOMER

Complaints

Customer complaints and warranty claims per thousand units sold ↓

These measures move before the euro figures do: they are leading indicators of the cost of quality.

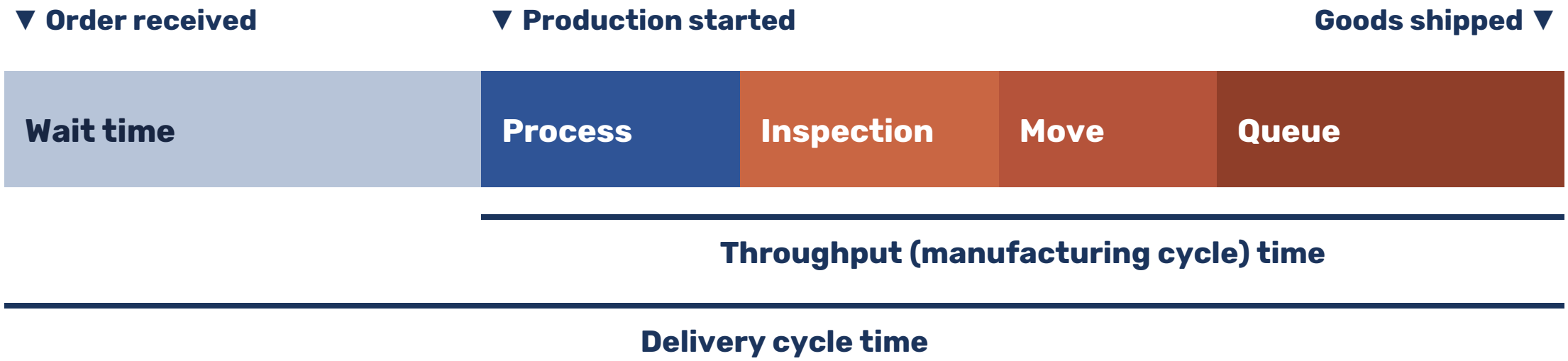
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Time is a performance measure

Section VII

The Balanced Scorecard

The life of a customer order



Value-added

Process time only: when work is actually done on the product.

Non-value-added

Wait, inspection, move and queue time. Reduce them as much as possible.

Throughput time (manufacturing cycle time)

The elapsed time from when production is started until finished goods are shipped to customers. Goal ↓.

Throughput time = Process + Inspection + Move + Queue

Process

Time when work is actually done on the product.

Inspection

Time spent making sure the product is not defective.

Move

Time to carry materials or partly finished products between workstations.

Queue

Time waiting to be worked on, moved, inspected or shipped.

Only one of the four activities adds value

ADDS VALUE

Process time

The customer would happily pay for this time: it transforms the product.

ADDS NO VALUE

Inspection, move, queue

The customer gains nothing from a product sitting in a queue, travelling across the plant or being checked. These should be eliminated as much as possible.

Inspection may feel useful, but it exists only because defects are possible. **Prevent the defects and inspection can shrink.**

Delivery cycle time

The elapsed time from when a customer order is **received** until the finished goods are **shipped**. This is the time the customer actually experiences. Goal ↓.

Delivery cycle time = Wait time + Throughput time

Wait time is the time from receiving the order until production starts. It adds no value. Companies that drastically cut wait time and the non-value-added parts of throughput time can raise both customer satisfaction and profit.

Manufacturing cycle efficiency (MCE)

MCE = Value-added time (process time) ÷ Throughput time

- MCE puts throughput time into perspective. Goal ↑, and the maximum is 1.
- Any non-value-added time makes MCE **less than 1**. An MCE of 0.5 means half the production time is spent inspecting, moving and waiting.

below 0.10

In many manufacturing companies MCE is below 10%: more than 90% of the time a unit is in process adds no value.

Novex Company: average times per order

Activity	Days
Wait time	17.0
Inspection time	0.4
Process time	2.0
Move time	0.6
Queue time	5.0

Goods are shipped as soon as production is completed. Compute:

1. Throughput time
2. Manufacturing cycle efficiency
3. Percentage of production time spent on non value added activities
4. Delivery cycle time

Solution

Four answers in four lines

8.0 days

Throughput
2.0 + 0.4 + 0.6 + 5.0

0.25

MCE
2.0 days ÷ 8.0 days

75%

Non-value-added share
100% − 25%

25.0 days

Delivery cycle
17.0 wait + 8.0 throughput

Once in production, a typical Novex order is actually being worked on only a quarter of the time. And the customer waits **17 days** before work even starts.

Quick check

Quick check 1: what is the throughput time?

Narton Corp. average times for production

Wait 3.0 days · Inspection 0.4 day · Process 0.2 day · Move 0.5 day · Queue 9.3 days

a 10.4 days

b 0.2 day

c 4.1 days

d 13.4 days

Quick check 2: what is the delivery cycle time?

Quick check

Narton Corp. average times for production

Wait 3.0 days · Inspection 0.4 day · Process 0.2 day · Move 0.5 day · Queue 9.3 days

a 0.5 day

b 0.7 day

c 13.4 days

d 10.4 days

Quick check

Quick check 3: what is the MCE?

Narton Corp. average times for production

Wait 3.0 days · Inspection 0.4 day · Process 0.2 day · Move 0.5 day · Queue 9.3 days

a 50.0%

b 1.9%

c 52.0%

d 5.1%

Solution

Quick check answers

Question	Working	Answer
1. Throughput time	Process + Inspection + Move + Queue = 0.2 + 0.4 + 0.5 + 9.3	a. 10.4 days
2. Delivery cycle time	Wait time + Throughput time = 3.0 + 10.4	c. 13.4 days
3. MCE	Process time ÷ Throughput time = 0.2 ÷ 10.4	b. 1.9%

Queue time is **9.3 of 10.4 days**: almost all of Narton's production time is spent waiting.

Codington Corporation, measured in hours

Activity	Hours
Wait time	20.8
Process time	2.2
Inspection time	0.2
Move time	4.8
Queue time	18.6

Throughput: 2.2 + 0.2 + 4.8 + 18.6 = 25.8 hours

MCE: 2.2 ÷ 25.8 = 0.09 (rounded)

Delivery cycle: 20.8 + 25.8 = 46.6 hours

Codington sits in the typical range: less than 10% of production time adds value.

What if Codington cuts waiting and queuing?

Scenario (hours)	Throughput	MCE	Delivery cycle
Base case	25.8	0.09	46.6
Wait 5.8, queue 8.6	15.8	0.14	21.6
Wait 1.0, move 0.8, queue 0.6	3.8	0.58	4.8

Why throughput and delivery fall

Queue time is part of both, and wait time is part of delivery cycle time. Removing non-value-added hours shortens both clocks.

Why MCE rises

Process time stays at 2.2 hours while throughput time shrinks, so the value-added share grows: 2.2 ÷ 3.8 = 0.58.

Your turn

Mittel wants to ship orders faster

First quarter	Days
Inspection time	0.3
Wait time	14.0
Process time	2.7
Move time	1.0
Queue time	5.0

Compute

1. Throughput time
2. MCE for the quarter
3. Percent of throughput time that is non value added
4. Delivery cycle time
5. New MCE if lean production removes all queue time

Solution

Removing queues more than doubles MCE

Measure	Working	Result
1. Throughput time	0.3 + 2.7 + 1.0 + 5.0	**9.0 days**
2. MCE	2.7 ÷ 9.0	**0.30**
3. Non value added	100% – 30%	**70%**
4. Delivery cycle time	14.0 + 9.0	**23.0 days**
5. MCE without queues	2.7 ÷ (0.3 + 2.7 + 1.0)	**0.675**

Eliminating 5 days of queue time cuts throughput time from 9.0 to 4.0 days. Delivery cycle time would fall to 18.0 days, but the 14 day wait before production would still dominate what the customer experiences.

Your turn

Four months of an automating plant

Measure	M1	M2	M3	M4
Number of defects	185	163	124	91
Warranty claims	46	39	30	27
Customer complaints	102	96	79	58
PO lead time, days	8	7	5	4
Scrap, % of cost	1%	1%	2%	3%
Downtime, % of availability	3%	4%	4%	6%
Use, % of availability	95%	92%	89%	85%
Setup time, hours	8	10	11	12
On time deliveries	96%	95%	92%	89%

Days	M1	M2	M3	M4
Wait	9.0	11.5	12.0	14.0
Inspection	0.8	0.7	0.7	0.7
Process	2.1	2.0	1.9	1.8
Queue	2.8	4.4	6.0	7.0
Move	0.3	0.4	0.4	0.5

1. Monthly throughput, MCE and delivery cycle time.

2. Where is Tombro improving, where deteriorating?

3. Month 4: throughput and MCE without queue time, then without queue and inspection.

Solution

Every time measure is getting worse

Days	Month 1	Month 2	Month 3	Month 4
Throughput time	6.0	7.5	9.0	10.0
MCE	0.35	0.27	0.21	0.18
Delivery cycle time	15.0	19.0	21.0	24.0

Month 1 working

Throughput 0.8 + 2.1 + 2.8 + 0.3 = 6.0

MCE 2.1 ÷ 6.0 = 0.35

Delivery 9.0 + 6.0 = 15.0

Requirement 3, from month 4

No queue: 0.7 + 1.8 + 0.5 = 3.0 days, MCE 0.60

No queue, no inspection: 1.8 + 0.5 = 2.3 days, MCE 0.78

Solution

Better quality, worse flow

Improving

- Defects: 185 to 91
- Warranty claims: 46 to 27
- Customer complaints: 102 to 58
- Purchase order lead time: 8 to 4 days
- Process time: 2.1 to 1.8 days

Deteriorating

- Scrap: 1% to 3% of cost
- Downtime up, machine use down, setups longer
- Throughput 6.0 to 10.0 days, MCE 0.35 to 0.18
- Delivery cycle 15 to 24 days
- On time deliveries: 96% to 89%

Quality is improving, but queues, waiting and setups are growing. Customers will feel the longer delivery times. **Without the time measures, management would only see the good news.**

Your turn

A new flexible plant, four months in

Average, days	Month 1	Month 2	Month 3	Month 4
Move time per unit	0.4	0.3	0.4	0.4
Process time per unit	2.1	2.0	1.9	1.8
Wait time per order	16.0	17.5	19.0	20.5
Queue time per unit	4.3	5.0	5.8	6.7
Inspection time per unit	0.6	0.7	0.7	0.6
On time deliveries	91%	86%	83%	79%
Total sales, units	3,210	3,072	2,915	2,806

Compute throughput, delivery cycle time and MCE for each month, evaluate the trend, then repeat requirement 3 of Tombro using month 4.

Solution

Faster processing, slower company

Days	Month 1	Month 2	Month 3	Month 4
Throughput time	7.4	8.0	8.8	9.5
Delivery cycle time	23.4	25.5	27.8	30.0
MCE	0.28	0.25	0.22	0.19

Evaluation

Process time is falling, but queue and wait times grow faster. Every time measure worsens, on time deliveries drop from 91% to 79% and unit sales fall each month.

Month 4, lean

Without queue time: 0.4 + 1.8 + 0.6 = 2.8 days, MCE 0.64.

Without queue and inspection: 0.4 + 1.8 = 2.2 days, MCE 0.82.

Time measures outside the factory

AMAZON

Delivery speed as a promise

Amazon tracks how many orders arrive the same day or the next day, and has reorganised its warehouses into regional networks to cut that time.

In 2024 it delivered **9 billion items** the same or the next day. Here the time measure is the value proposition itself.

ZARA (INDITEX)

From design to shop in weeks

Traditional fashion plans its collections many months ahead. Zara designs, produces and ships in short cycles, much of it close to Spain, and can bring a new design to its shops in a few weeks.

Shorter throughput time means **less unsold stock** and fewer end of season discounts.



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Overall equipment effectiveness

Section VIII

The Balanced Scorecard

Three ways a machine can disappoint us

Overall equipment effectiveness measures the productivity of a piece of equipment along three dimensions. It is widely used by lean manufacturers. Goal ↑.

Utilisation

Is the machine running when it could be? Breakdowns and long setups leave it idle.

Efficiency

When it runs, does it run at full speed? Minor stoppages slow it down.

Quality

Of what it produces, how much is free of defects?

OEE is the product of three rates

OEE = Utilisation rate × Efficiency rate × Quality rate

Rate	Formula
Utilisation rate	Actual run time ÷ Machine time available
Efficiency rate	Actual run rate ÷ Ideal run rate
Quality rate	Defect free output ÷ Total output

Each rate is between 0 and 1, so OEE is always smaller than the weakest of the three. Three rates of 90% give an OEE of only **73%**.

How to raise each rate

Rate	What managers can do
Utilisation	Avoid machine breakdowns and minimise average setup times.
Efficiency	Avoid minor work stoppages; retain and properly train operators to run and regularly maintain the machine.
Quality	Reduce the number of defective units as a percent of total output.

Notice the learning and growth link: retaining and training operators is a people measure that should drive a process measure. A ready made **if/then** hypothesis.

Jayson Company's bottleneck machine

This week	Value
Actual run time	4,550 minutes
Machine time available	6,500 minutes
Actual run rate	3.8 units/minute
Ideal run rate	4.0 units/minute
Defect free output	16,000 units
Total output, including defects	17,290 units

Calculate the machine's OEE

Demand for Jayson's only product is higher than its capacity.

This machine is the constraint: every unit it fails to make is a unit Jayson cannot sell.

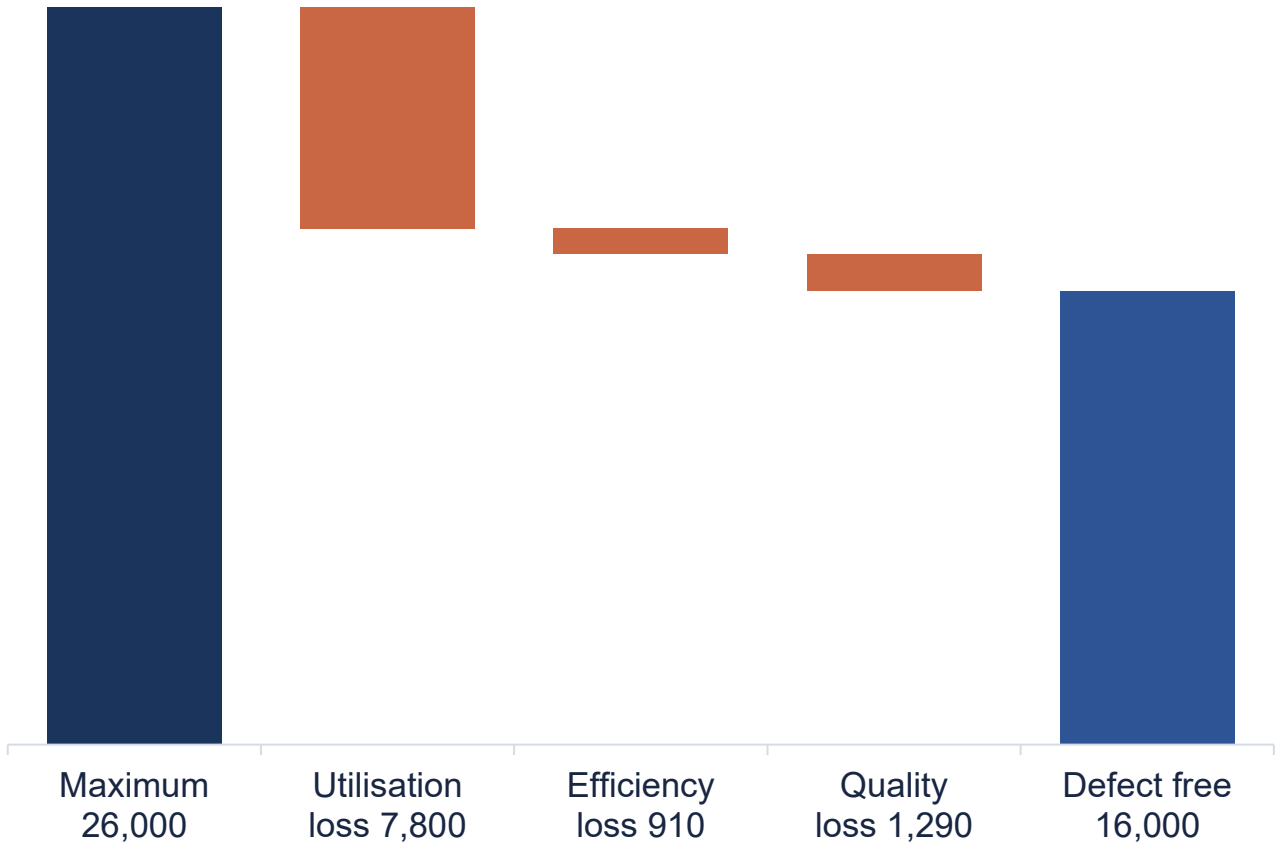
Solution

OEE of 0.615

UTILISATION		EFFICIENCY		QUALITY		OEE
0.70	×	**0.95**	×	**0.925**	=	**0.615**
4,550 ÷ 6,500		3.8 ÷ 4.0		16,000 ÷ 17,290		of maximum output

The weakest rate is utilisation: the machine stood idle for **1,950** of its 6,500 available minutes. Breakdowns and setups are the first place to look.

Jayson lost 10,000 units it could have sold



Start from what the machine could make: **26,000 units**.

Three losses take it down:

- idle time: (7,800)
- running slower than ideal: (910)
- defective units: (1,290)

What is left: **16,000 good units**, and 16,000 ÷ 26,000 = **0.615**, the same OEE.

10,000 products that customers wanted and did not get.

How each number in the chart was calculated

Step	What it captures	Formula	Jayson	Units
1. Maximum output	Machine never stops, slows or errs	Time available × Ideal rate	6,500 × 4.0	26,000
2. Utilisation loss	Minutes the machine stood idle	(Time available − Run time) × Ideal rate	(6,500 − 4,550) × 4.0 = 1,950 × 4.0	(7,800)
3. Efficiency loss	Running slower than ideal	Run time × (Ideal rate − Actual rate)	4,550 × (4.0 − 3.8) = 4,550 × 0.2	(910)
4. Quality loss	Units made but defective	Total output − Defect free output	17,290 − 16,000	(1,290)
Defect free output				**16,000**

Check: 26,000 − 7,800 − 910 − 1,290 = 16,000 units. OEE in units: 16,000 ÷ 26,000 = 0.615, the same as 0.70 × 0.95 × 0.925.

Your turn

Another example: Kilmer

This week	Value
Actual run time	4,900 minutes
Machine time available	7,000 minutes
Actual run rate	3.8 units/minute
Ideal run rate	4.0 units/minute
Defect free output	15,827 units
Total output	18,620 units

Compute

1. The utilisation, efficiency and quality rates, and the OEE.

2. Maximum output and the three losses in units, following the same steps as Jayson.

Hint: maximum output = 7,000 × 4.0.

Solution

Kilmer, step by step

UTILISATION	×	EFFICIENCY	×	QUALITY	=	OEE
4,900 ÷ 7,000		3.8 ÷ 4.0		15,827 ÷ 18,620		0.70 × 0.95 × 0.85
0.70		**0.95**		**0.85**		**0.565**

Step	Formula	Kilmer	Units
1. Maximum output	Time available × Ideal rate	7,000 × 4.0	28,000
2. Utilisation loss	(Time available − Run time) × Ideal rate	(7,000 − 4,900) × 4.0 = 2,100 × 4.0	(8,400)
3. Efficiency loss	Run time × (Ideal rate − Actual rate)	4,900 × (4.0 − 3.8) = 4,900 × 0.2	(980)
4. Quality loss	Total output − Defect free output	18,620 − 15,827	(2,793)
Defect free output			**15,827**

Check: 28,000 − 8,400 − 980 − 2,793 = 15,827, and 15,827 ÷ 28,000 = 0.565.

Your turn

Rates and lost units for Kilmer

This week	Value
Actual run time	6,600 minutes
Machine time available	11,000 minutes
Actual run rate	5.1 units/minute
Ideal run rate	6.0 units/minute
Defect free output	23,562 units
Total output	33,660 units

Compute

1. Utilisation rate, efficiency rate, quality rate and OEE.

2. Utilisation loss, efficiency loss and quality loss in units for the week.

Hint: start from the maximum possible output, 11,000 × 6.0.

Solution

An OEE of 0.357: almost two thirds lost

Rate	Working	Result
Utilisation	6,600 ÷ 11,000	0.60
Efficiency	5.1 ÷ 6.0	0.85
Quality	23,562 ÷ 33,660	0.70
OEE	**0.60 × 0.85 × 0.70**	**0.357**

Units	Working	Result
Maximum	11,000 × 6.0	66,000
Utilisation loss	4,400 × 6.0	26,400
Efficiency loss	0.9 × 6,600	5,940
Quality loss	33,660 − 23,562	10,098
Good output	**66,000 − 42,438**	**23,562**

Check: 23,562 ÷ 66,000 = 0.357. Idle time is the biggest loss, but Kilmer also scraps three units in ten. Both problems deserve attention **before buying a new machine**.



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Building a scorecard

Section IX

The Balanced Scorecard

Four steps to a scorecard

1

Define the strategy

Name the target customers and the value proposition.

2

Choose measures

A few per perspective, each with a direction.

3

Link them in a strategy map

Draw arrows and write each arrow as an if/then statement.

4

Test and reward

Track the data, check the links, and only later tie pay to the measures.

In brief: define the strategy, translate it into measures that track its execution, and link those measures in a **cause and effect** manner.

Your turn

Which value proposition is Jaguar pursuing?

“Jaguar's strategy is to offer distinctive, luxury automobiles to wealthy individuals who prize handcrafted, individualised products.”

Customer intimacy

?

Operational excellence

?

Product leadership

?

Which words in the strategy statement give the answer away?

A strategy for Jaguar

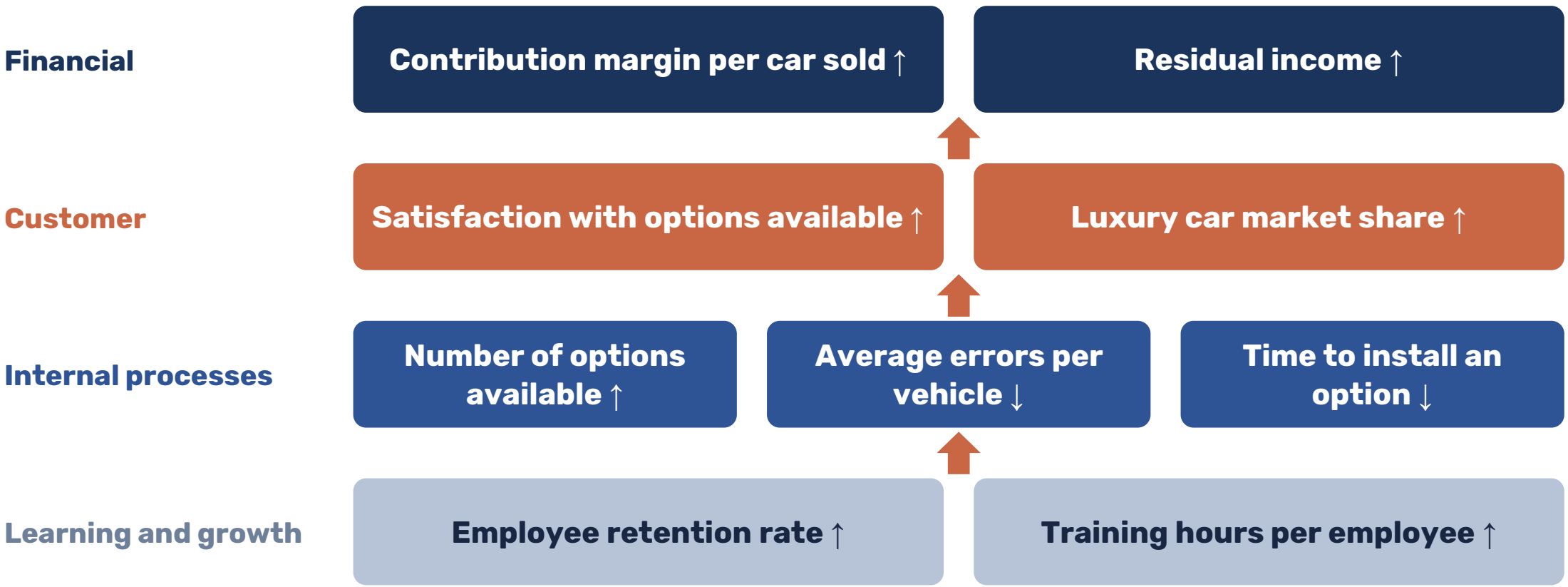
- Suppose Jaguar offers distinctive, richly finished luxury cars to wealthy buyers who prize handcrafted, individual products. This is a **customer intimacy** proposition.
- So many options for leather, colours and wood dashboards that each car is almost one of a kind: not tan or blue cowhide, but an almost unlimited palette of exotic leathers.

The catch

Jaguar must deliver a fully customised car in a reasonable time, and without spending more on customisation than the customer is willing to pay.

Strategy map

A possible balanced scorecard for Jaguar



Reading Jaguar's arrows as hypotheses

- **If** retention and training hours go up, **then** options available go up, while errors per vehicle and time to install an option go down.
- **If** options go up and errors and install time go down, **then** customer satisfaction with options goes up, and with it luxury car market share.
- **If** satisfaction with options goes up and install time goes down, **then** contribution margin per car goes up.
- **If** market share and contribution margin per car go up, **then** residual income goes up.

THE SCORECARD IS A THEORY

It lays out how concrete actions should lead to the results we want. Data then tells us whether the theory holds.

Suppose Jaguar offers more options, makes fewer errors and installs faster, yet satisfaction, share, margin and residual income do not move. Then the strategy must be reconsidered.

Without this feedback a company may drift on for years with an ineffective strategy built on faulty assumptions.

What makes a good if/then statement

WEAK

“If employees are happier, the company will do better”.

Neither side is a measure, there is no direction, and nothing can be checked.

STRONG

“If employee turnover ↓, then the number of warranty claims ↓”.

Two named measures, a direction for each, and a mechanism we can explain: experienced staff make fewer mistakes.

Checklist: two specific measures · a direction for each · usually one perspective to the next · a plausible reason · data you can actually collect.

Your turn

Write the hypothesis, then break it

	If	Then
1	Employee turnover percentage	Number of warranty claims
2	Customer defection rate	Sales growth rate
3	Number of new products designed	Share of wallet
4	Non value added activity costs	Net operating income
5	Process improvement suggestions per employee	Non value added activity costs

1. Write each row as an if/then statement with a direction for both measures. **2.** Give one reason why improving the “if” measure might **not** improve the “then” measure.

Solution

Five hypotheses and how they could fail

Hypothesis	Why the link might not hold
1. If turnover ↓, then warranty claims ↓	Defects may come from poor designs or supplier parts, not from inexperienced staff.
2. If defection rate ↓, then sales growth ↑	Loyal customers may buy less each year, or prices may have been cut to keep them.
3. If new products designed ↑, then share of wallet ↑	The new products may not match customer needs, or may simply replace sales of older ones.
4. If non value added costs ↓, then net operating income ↑	Freed capacity may sit idle, or the costs are fixed and do not actually disappear.
5. If suggestions per employee ↑, then non value added costs ↓	Suggestions may be of poor quality or never put into practice.

Your turn

A scorecard for the Powder 8 Lodge

A small family resort for day skiers has been bought by Western Resorts, which wants a destination resort. The cafeteria serves only burgers, hot dogs, chili, sandwiches, pizza, fries and snacks.

All staff were kept. The manager is enthusiastic but new to restaurants. Kitchen staff cook and wash up; dining room staff take orders, act as cashiers and clean.

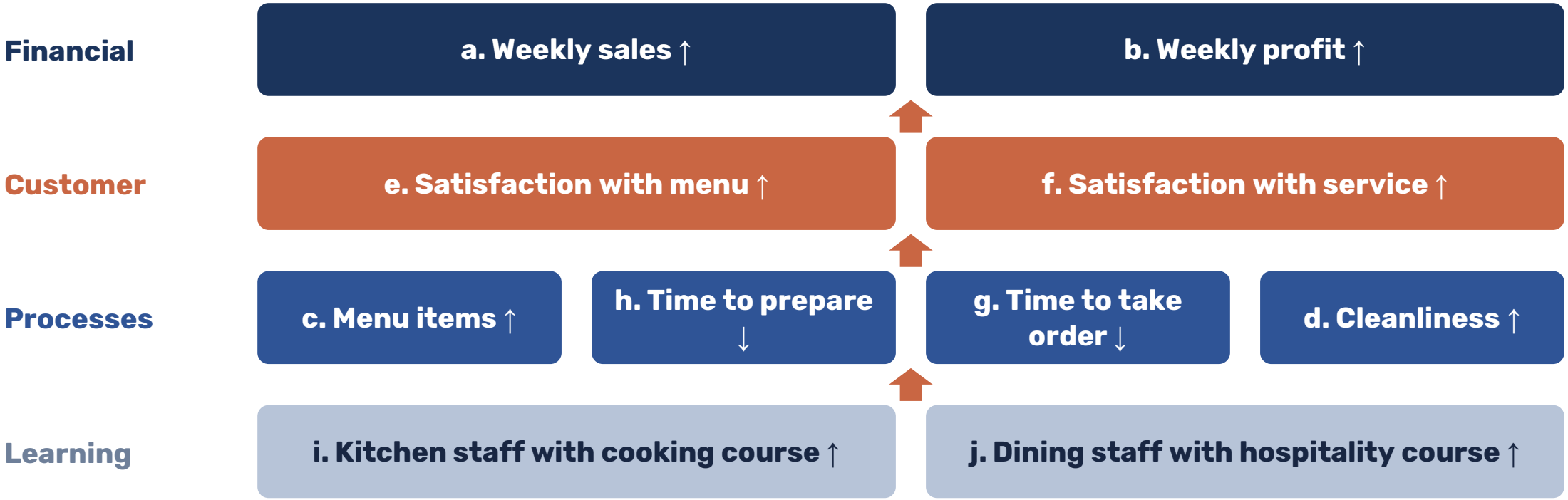
1. Draw the scorecard with arrows and ↑ or ↓. **2.** Which hypotheses are most questionable? **3.** How will management know one is false?

Measures chosen

a. Weekly lodge sales
b. Weekly lodge profit
c. Number of menu items
d. Dining area cleanliness rating
e. Satisfaction with menu choices
f. Satisfaction with service
g. Average time to take an order
h. Average time to prepare an order
i. % kitchen staff with a cooking course
j. % dining staff with a hospitality course

Solution

One possible scorecard



Links: i drives c and h; j drives g and d. c drives e; h, g and d drive f. e and f drive a; a drives b.

Which links would you doubt?

Courses lead to better work

A college course may not change daily habits in a busy ski cafeteria, especially without an experienced manager.

More menu items raise satisfaction

Skiers may want speed more than choice. A longer menu could also slow preparation.

Satisfaction raises sales

The lodge is the only place to eat on the hill. Sales may depend more on how many skiers come.

Sales raise profit

Not if the new menu and training cost more than the extra sales bring in.

How management will know: track every measure over time. If an “if” measure improves but its “then” measure does not, that link is in doubt.

Your turn

Two problems, sixteen candidate measures

A downtown store losing ground to national chains has two problems: customers pay their store card bills late and bad debts are high, mostly because clerks enter wrong data and bills are disputed; and unsold seasonal clothing is sold off at a loss.

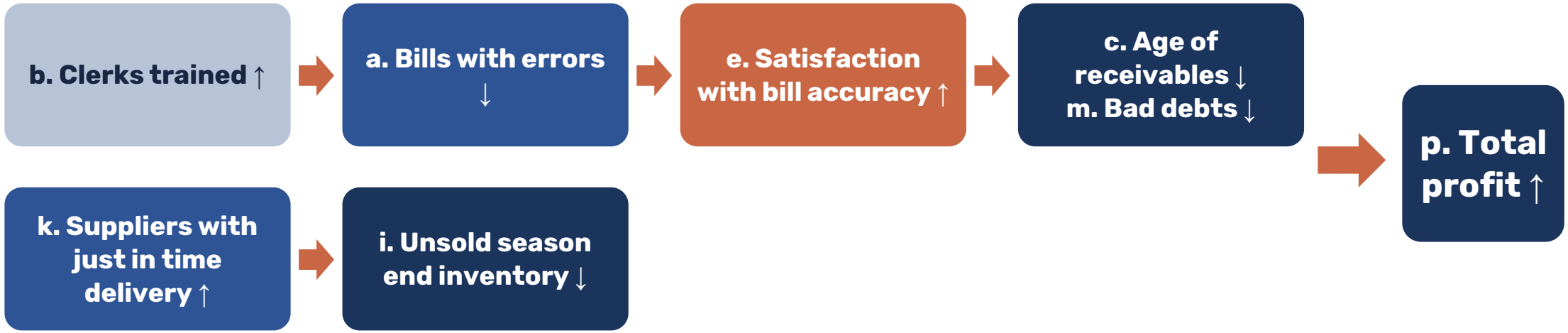
a. % of bills containing errors
b. % of clerks trained to enter data correctly
c. Average age of receivables
d. Profit per employee
e. Satisfaction with bill accuracy
f. Total sales revenue
g. Sales per employee
h. Buyers' travel to fashion shows

i. Unsold season end inventory, % of cost of sales
j. Courtesy of junior to senior staff
k. % of suppliers with just in time delivery
l. Sales per square foot
m. Bad debts as % of sales
n. Quality of staff cafeteria food
o. % attending a diversity workshop
p. Total profit

Task: using only these measures, build a linked scorecard that shows a strategy for both problems. You need not use them all.

Solution

Two chains that meet in profit



Why: trained clerks make fewer errors, accurate bills are not disputed, so customers pay sooner and fewer debts go bad. Suppliers who deliver just in time let buyers order smaller quantities closer to the season, so less stock is left over. Measures d, f, g, j, l, n and o address neither problem.

When some measures move and others do not

After a year, mixed results

Find the first link in each chain that failed. Is the hypothesis wrong, or does the effect need more time? Then revise the measures or the strategy.

Accurate bills, still late payers

Billing errors were not the main cause. Credit may be granted too easily, collection may be weak, or customers may be short of money.

Better numbers, same profit

Savings may be eaten by training or supplier costs. Leaner stock may cause lost sales; tighter credit may push customers to the chains.

This is the scorecard doing its job: it does not only report a disappointing profit, it points to **which assumption** to examine.



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Using the scorecard

Section X

The Balanced Scorecard

Tie pay to the scorecard, but not on day one

- Bonuses can, and probably should, be linked to scorecard measures. Otherwise the measures are likely to be ignored.
- But only after the organisation has been managed with the scorecard for some time, **perhaps a year or more**.

First be sure the measures are

- reliable
- sensible
- understood by the people evaluated
- not easily manipulated

Pipeline Unlimited's commission plan

A surfing equipment maker runs at 75% of capacity. It puts **sales growth** on its scorecard and pays salespeople **10% of sales revenue**. Now look at two surfboards.

Per board	XR7	Turbo
Selling price	$695	$749
Variable expenses	344	410
Contribution margin	**$351**	**$339**
Commission at 10% of sales	$69.50	**$74.90**

Which board will salespeople push? **The Turbo.** Which board earns the company more? **The XR7.**

Pay on contribution margin instead

10% OF SALES

Salespeople push the Turbo

XR7: $69.50

Turbo: **$74.90**

The incentive points to the less profitable board.

10% OF CONTRIBUTION MARGIN

Salespeople push the XR7

XR7: **$35.10**

Turbo: $33.90

What is best for the salesperson is now best for the company.

As long as fixed costs do not depend on the sales mix, maximising contribution margin also maximises profit. A good scorecard creates the same alignment: people who maximise their own reward also advance the strategy.

Your turn

How could a manager game each measure?

1. Speed to market

An electronics maker holds R&D responsible for the average time a product spends in development before release.

2. Luggage unloading

An airline pays ground crews a bonus each month their average time from parking to last bag unloaded beats the previous month.

3. On time shipments

The production manager agrees a delivery date with each customer and is judged on the percent of orders shipped by that date.

4. Revenue per employee

Subsidiary managers of a multinational are held responsible for increasing revenue per employee.

Solution

The number improves, the company does not

Measure	The trick	Why the number improves but the company does not	Counterweight
1. Speed to market	Release products before they are fully tested	Development time falls, but defects and returns rise after launch	Defect rate after launch
2. Luggage unloading	Rush the bags, or work slowly one month and fast the next	The bonus rewards beating last month, so a slow month sets up an easy win	Damaged bags; a fixed target
3. On time shipments	Agree delivery dates far in the future	The share of orders shipped on time rises, while customers wait longer	Delivery cycle time
4. Revenue per employee	Replace employees with contractors, or cut prices	Fewer heads or more volume raise the ratio; costs rise and margins fall	Profit per employee

Lesson: a single measure invites gaming. Pair it with a counterweight that gets worse when someone cheats.

Wells Fargo: when one measure ruled the bank

- Wells Fargo, one of the largest US banks, built its strategy on **cross selling**: the number of products each customer holds. The internal motto was “eight is great”.
- Branch employees faced daily sales targets. To meet them, they opened up to **3.5 million** accounts that customers had never asked for.
- In 2020 the bank agreed to pay **$3 billion** to settle criminal and civil investigations, and its reputation suffered for years.

THE LESSON

What the scorecard was missing

The measure

Products per customer ↑

The counterweights

- Accounts actually used ↑
- Customer complaints ↓
- Branch employee turnover ↓

Beyond shareholders

- Companies must produce financial results that satisfy shareholders. They also have a responsibility to other stakeholders whose interests are tied to the company's performance.
- **Corporate social responsibility (CSR)** means considering the needs of all stakeholders when making decisions. It goes beyond legal compliance to include voluntary actions.

Stakeholders

- Customers
- Employees
- Suppliers
- Communities
- Environmental advocates
- Human rights advocates

Opportunity and risk

Opportunity

Toyota developed the hybrid Prius to answer emission concerns and grow shareholder wealth.

Timberland makes Earthkeepers shoes from recycled plastic bottles; the line grew fast and profitably.

Procter & Gamble merged distribution centres, cutting both costs and emissions.

Risk

Polluting, buying from suppliers that use child labour, or discriminating against employees damages reputation.

The after effects: boycotts, fewer strong job applicants, higher turnover, unhappy investors, lawsuits and fines.

Whatever the motive, morality, opportunity or risk, the interest needs measures that managers can track and stakeholders can read.

The GRI framework

The GRI is a leading organisation in social and environmental performance measurement. Its widely adopted sustainability reporting framework has **universal standards** and three sets of **topic specific standards**.

Universal

Senior leader's statement on CSR; stakeholders and how often we talk to them; the board's role in risk and reporting.

Economic

Retirement funding, local suppliers and managers, entry wages by gender, climate risk.

Environmental

Water, waste, energy, biodiversity and emissions.

Social

Health and safety, human rights, training, diversity, product safety, labelling.

The three topic areas mirror the “triple bottom line”: people, planet and profit.

CSR measures in the four perspectives

Financial

Revenue from eco conscious products ↑
Fuel cost per sales dollar ↓
Total workers' compensation costs
Capital spending on green products and processes

Customer

Customers who strongly agree that we care about communities ↑, product safety ↑, environmental stewardship ↑, and inform them about our CSR performance ↑

Internal processes

Water consumed per unit ↓
Recycled raw material inputs ↑
Carbon emissions per unit ↓
Pounds of waste ↓

Learning and growth

Occupational injuries ↓
Female to male average pay ↑
Applicants from under represented groups ↑
Community service hours per employee ↑

These measures matter only when linked to strategic goals and to rewards.

Three ways to bring CSR into the scorecard

Inside the four boxes

Add social and environmental measures to the existing perspectives.

A fifth perspective

Add a category dedicated to a wider set of stakeholders.

A separate scorecard

Build a dedicated scorecard for the company's own, or its suppliers', social and environmental performance.

Whichever format is chosen, two rules hold: link the measures with if/then statements, and tie them to how senior managers are evaluated and rewarded. **Otherwise a proclaimed commitment may not lead to action.**

What we take away, and what comes next

1. **Strategy first:** the value proposition decides which measures go on the scorecard.
2. **Four perspectives, linked** by if/then hypotheses and drawn as a strategy map.
3. **Non-financial measures** such as quality, time and OEE lead the financial results.
4. **Balance and rewards:** pair each measure with a counterweight, then link it to pay.

NEXT LECTURE

The Sustainable Balanced Scorecard

When stakeholders include the environment and society, the map needs new objectives.

Think about it before next time

- Where does Enel's renewable capacity belong on its strategy map?
- Which environmental measure would you add to Ryanair's map?